Can International Initiatives Promote Peace?

Diamond Certification and Armed Conflicts in Africa

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Abstract

The Kimberley Process Certification Scheme (KPCS) aims to prevent so-called conflict diamonds – diamonds that originate from conflict zones – from entering the world market. This paper examines its impact on armed conflict in Africa by exploiting grid-cell level variation in the suitability to extract alluvial (secondary) diamonds and comparing grid cells with and without this suitability before and after the introduction of the KPCS in 2002. We find that the KPCS led to a permanent and significant reduction in armed conflict without triggering increases in other forms of violence. We also provide evidence for geographic spillovers: conflict also declined in neighboring regions. Rebel group-level evidence further shows that after the introduction of the KPCS, rebel groups were less likely to engage in armed conflict and their operations became more geographically concentrated. Overall, the findings suggest that the KPCS effectively limited rebels' ability to finance violent activities.

Keywords: Armed conflicts, diamonds, Africa, certification schemes, trade restrictions

JEL: D74, F55, F63, O55, Q34

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1 Introduction

Over the past few decades, there has been a rise in international agreements, national regulations, and transparency and certification schemes that aim to address global challenges, such as poverty and vulnerability, climate change, and violent conflicts (e.g. Dragusanu et al., 2014; Christensen et al., 2021; Rohner, 2024).[1] The success of such efforts has been subject to considerable controversy, in part due to a lack of rigorous empirical evidence (Berman et al., 2017; Oya et al., 2017). One recurring concern is that weak governance structures and high levels of corruption may undermine the effectiveness of initiatives to ameliorate social and environmental problems in fragile states. At the same time, these states are a prime target for many initiatives, as efforts to build state capacity and promote economic development have proven difficult (Page and Pande, 2018; Bandiera et al., 2019).

We examine one of the first international initiatives aimed at preventing the illicit exploitation of natural resources and promoting peace: the Kimberley Process Certification Scheme (KPCS). By tracking diamonds and limiting trade to diamonds that are certified, the KPCS aims to prevent so-called conflict diamonds – "rough diamonds used to finance wars against governments"[2] – from entering world markets (e.g. Haufler, 2009). To study its impact on armed conflicts in Africa, we compare grid cells with and without suitability for extracting alluvial (secondary) diamonds over time. We document a permanent and significant decline in armed conflict after the introduction of the KPCS, along with reduced rebel group activities concentrated in fewer geographic areas. As other forms of violence did not increase, our findings suggest that the KPCS effectively limited rebels' ability to finance violent activities.

The Kimberley Process was initiated after civil society organizations and the UN called attention to the role of "conflict diamonds" (AKA "blood diamonds") in financing brutal wars in such states as the Democratic Republic of the Congo, Angola, and Sierra Leone in the late 1990s (Haufler, 2009; Bieri, 2015). Conflict diamonds generally constitute alluvial, or secondary, diamonds that are extracted across Africa by individual diggers using simple tools.[3] Exploiting commodity price changes, past research has shown that alluvial minerals, including diamonds and gold, increase the incidence of violent conflict in Africa (Rigterink, 2020; Blair et al., 2021; Rigterink et al., 2025).

While the KPCS led to a significant increase in official diamond exports in producing countries (Haufler, 2009), its impact on armed conflict in Africa has remained unclear. Stakeholders of the process and case studies have highlighted both achievements and limitations of the KPCS (Smillie, 2005; Haufler, 2009; Smillie, 2014; Bieri, 2015; Engwicht, 2018). For example, critics raise concerns that the KPCS relies on member states' willingness and ability to

[1] For example, according to the International Trade Center's "Standards Map," there are over 300 voluntary sustainability standards alone (https://www.standardsmap.org/).

[2] This is the official definition of conflict diamonds used by the KPCS (link).

[3] Alluvial mining of minerals is widespread in Africa and has increased considerably over the past decades (World Bank, 2019; Girard et al., 2025). In 2017, there were some 10 million artisanal and small-scale miners in Africa, a figure equivalent to 2.4% of the continent's labor force (World Bank, 2019, p.12).

monitor and enforce the scheme and that the possibility of disciplining participants is limited. Empirically, there exists only suggestive evidence, primarily due to previous data limitations, with Heffernan (2016) observing a conflict-reducing effect and Berman et al. (2017) finding no substantial effect.[4]

To study the causal impact of the KPCS on armed conflicts in Africa, we exploit cross-sectional variation in the suitability for secondary diamond mining within 0.5×0.5 degree grid cells and compare the incidence of armed conflict in grid cells with and without secondary diamond (SD) suitability over the period 1997 to 2013, accounting for grid cell and region $\times$ year fixed effects and a set of time-varying controls. We use data on secondary diamond suitability from Rigterink (2020) rather than actual mining activities as the latter is likely endogenous to armed conflict and measured with error.[5] Geo-referenced data on armed conflict ("battles") comes from the Armed Conflict Location & Event Data Project (ACLED) and the UCDP/PRIO Armed Conflict Dataset (UCDP). The underlying assumption is that the KPCS's impact on armed conflicts is concentrated on grid cells suitable for SD mining. This seems plausible, as the KPCS aims to make areas with alluvial diamonds less attractive to rebel groups by shutting down options for rebel groups to sell rough diamonds in the formal market. It is also supported by empirical evidence that we present below.

Our results document a permanent decline in the incidence of armed conflict in secondary diamond suitability grid cells after 2001, when members began implementing the KPCS. As we consider diamond suitability rather than actual diamond mining activities and as we later document a decline in armed conflict also in neighboring grid cells, our estimates can be seen as a lower bound for the true effect. Our estimates based on ACLED and UCDP suggest that the KPCS reduced the incidence of armed conflict by about half in the first 12 years post-KPCS.

Our difference-in-differences approach requires that grid cells with and without SD suitability would have experienced similar changes in armed conflict after 2001 if the KPCS had not been introduced. We demonstrate that divergent trends in armed conflict emerged only after 2001, not before. In addition, we run a number of robustness analyses to address concerns about our identification strategy. First, we demonstrate that our finding remains robust when excluding grid cells located in Angola and Sierra Leone, whose wars ended in 2002. We also examine the sensitivity of our estimates to dropping other single countries and groups of countries. Second, during the time period we consider, several transparency and reporting schemes were implemented that targeted firms engaged in large-scale industrial mining (Berman et al., 2017; Christensen et al., 2024). To rule out that our estimates capture

[4] Heffernan (2016) studies the impact of the KPCS on armed conflict at the country-level by comparing countries with secondary diamond deposits with a synthetic control group. In the context of Africa, Berman et al. (2017) analyze the relationship between mineral prices and conflict, focusing on industrial rather than alluvial mining. They note, however, that the use of diamond prices is problematic due to considerable fluctuations in the quality and type of diamonds.

[5] Our approach is similar in spirit to analyses using agricultural suitability rather than agricultural productivity. Early work includes Nunn and Qian (2011) and Alesina et al. (2013). Following these papers, we propose using the term suitability instead of propensity, the term used by Rigterink (2020) when introducing her data.

in part the impact of these schemes, we make use of the fact that primary diamonds can be extracted from the host rock only, while alluvial diamonds can also be extracted far away from it, along downstream river segments (for details see Section 2). Reassuringly, we obtain similar estimates when removing grid cells with a suitability for primary diamonds. Third, Berman et al. (2017) document that the rise in world mineral prices over the 1997–2010 period led to an increase in armed conflict in places with industrial mining. We, therefore, modify our baseline specification in two ways: we add interactions between a grid cell's distance to an industrial diamond mine and year fixed effects, and we add interactions between SD suitability grid cells and the world price of each of the five major minerals in Africa. The estimates are slightly smaller in both regressions, but remain statistically and economically significant. Fourth, we increase the comparability between the treatment and the control groups by restricting the sample to grid cells on or near a major river and, alternatively, to grid cells with alluvial gold suitability; for alluvial gold suitability, we use newly available data from Girard et al. (2025). Our result is robust to both exercises, although the coefficient estimate is smaller in the first exercise. Finally, we use ACLED to show that our result is not sensitive to the measure of armed conflict, and we document that violence against civilians as well as overall violence, which includes battles, violence against civilians, and explosions/remote violence, also declined.

The various robustness analyses thus support a causal relationship between the KPCS and armed conflict. We perform two additional exercises to validate our identification strategy. First, in a placebo test, we demonstrate that the KPCS does not reduce conflict incidence related to non-diamond artisanal and small-scale mining (ASM) activities. To this end, we compare grid cells with and without alluvial gold suitability over time, as alluvial gold mining constitutes the main form of alluvial mining in Africa (Girard et al., 2025). We find no differential trend in the incidence of armed conflict in relation to a grid cell's gold suitability, neither before nor after the introduction of the KPCS. Second, we exploit variation in treatment intensity, i.e. variation within the group of SD suitability grid cells. We demonstrate that the impact is greater for SD suitability grid cells located in early KP member countries (compared to late or non-KP member countries) and for grid cells with high SD suitability (compared to those with low suitability). Finally, we document that, as intended, it is mainly countries with weak political institutions – as defined by the Polity5 Democracy Index or by the Worldwide Governance Indicators – that benefit from the introduction of the KPCS.

Our findings are consistent with the KPCS's core idea of breaking the link between natural resources and conflict. By introducing barriers to diamond trading and institutional reforms in member countries, the KPCS aims to restrict access to diamond revenues, thereby directly decreasing the financial capacity of rebel groups and reducing their feasibility of initiating, sustaining, and escalating violent activities. Consequently, with the introduction of the KPCS, rebel groups faced pressure to relocate their activities to other areas or shift their business model to different activities and revenue sources, potentially leading to other forms of violence. We follow two complementary strategies to provide empirical evidence on this issue.

First, we investigate whether rebel groups relocate their activities away from SD suitability grid cells to neighboring grid cells, thereby increasing violence (in the form of armed conflict or violence more broadly) in these areas. We find evidence for positive geographic spillovers. That is, the KPCS reduced armed conflict and other forms of violence not only in grid cells with SD suitability but also in grid cells located in close proximity to these cells. One caveat of this analysis is that rebel groups may have relocated their violent activities to places further away from SD suitability grid cells. We therefore next consider changes in rebel group activity over time using data at the rebel group-level from ACLED. We limit the analysis to rebel groups that were involved in armed conflicts before the introduction of the KPCS. We then compare rebel groups with armed conflict activities in SD suitability grid cells prior to the KPCS with rebel groups without such activities. We document that following the introduction of the KPCS, rebel groups engage in fewer armed conflicts, and their armed conflict activities become more geographically limited. This holds when considering only armed activities in grid cells without SD suitability, as well as for violence against civilians and for violent conflicts overall. These various results suggest that the KPCS was successful in the intended manner: it restricted a core income source of rebel groups, leading to an overall decline in violent conflict.

Our paper contributes to several strands in the literature. There is a rich literature on the determinants of violent conflict, particularly studies focusing on Africa (e.g. Michalopoulos and Papaioannou, 2016; Harari and Ferrara, 2018; McGuirk and Burke, 2020; McGuirk and Nunn, 2024; Eberle et al., 2025). Some studies have examined the link between natural resources and conflict, both theoretically and empirically, with more recent work drawing on grid cell level data (Berman et al., 2017; Rigterink, 2020; Bhattacharyya and Mamo, 2021; Blair et al., 2021; Hodler et al., 2023; Rigterink et al., 2025; and for a recent review Rohner, 2025).We contribute to this literature by documenting that policies have the potential to reduce violent conflicts in fragile states and thus mitigate what has been termed the "natural resource curse."

Our paper also ties into the nascent literature that tries to identify the causal impact of national or international initiatives, including Corporate Social Responsibility (CSR) and sustainability reporting on mitigating conflict and/or improving the well-being of local communities in developing countries (e.g. Berman et al., 2017; Dragusanu et al., 2022; Baik et al., 2024; Christensen et al., 2024). Berman et al. (2017), focusing on industrial mining, provide some initial evidence that various transparency initiatives may have reduced the link between price spikes and violent conflict in Africa. More recent work, also drawing on fine-grained data, exploits policy changes over time in a difference-in-differences framework. Christensen et al. (2024) document an increase in economic development in African communities near large resource extraction facilities following a major increase in US Foreign Corruption Regulation enforcement after 2004. Baik et al. (2024) show that after 2014, when the introduction of the US conflict minerals disclosures rule (Section 1502 of the Dodd–Frank Act) went into effect, conflict incidence declined in African regions covered by the rule.[6] In a more detailed exami-

[6] Earlier work on the regulation points to an increase in conflicts (Parker and Vadheim, 2017; Stoop et al.,

nation of the role of certification for conflict-free mines in the Eastern Democratic Republic of the Congo, Chang and Christensen (2023) find a decrease in conflicts after certification, but also evidence for a displacement of conflicts to nearby areas. We contribute to this literature by examining a scheme that focuses on alluvial diamond mining, which is difficult to regulate. In contrast to Chang and Christensen (2023), we find no evidence for displacement effects. As the KPCS requires at least some involvement by those countries engaged in the production of alluvial diamonds, our study is also related to work on the importance of domestic regulation in addressing social and environmental issues as countries progress (Jayachandran, 2022) and to a vast literature on corruption in low- and middle-income countries, including studies on interventions by governments to curb corruption (for reviews see Olken and Pande, 2012; Finan et al., 2017). We contribute to this literature by studying a certification scheme that combines domestic and international efforts to reduce illicit trade in alluvial diamonds.

Finally, and more broadly, our paper is related to the literature that examines whether and how international interventions such as development aid, trade restrictions, and sanctions can promote peace (e.g. Crost et al., 2014; Nunn and Qian, 2014; Moscona, 2025; for reviews see Hoeffler, 2014; Findley, 2018; Morgan et al., 2023; McGuirk and Trebesch, 2025).

2 Background

Diamonds are either excavated from their host rock through large, capital-intensive deep-mining operations ("primary diamonds") or they are extracted through labor-intensive, alluvial surface mining operations ("secondary diamonds"). The latter can be found along rivers many kilometers away from their host rock. They are extracted across Africa by individual diggers using simple tools. In Africa, the first alluvial diamonds and primary diamonds (kimberlite pipes) were discovered in South Africa, near Kimberley, in the late 1860s and 70s (Janse, 2007). Over the next decades, further diamond deposits were discovered in South and Central Africa and, later, in the 1930s and 40s, in East and West Africa (Janse, 1995a,b). In 2000, African countries with mostly deep mining were South Africa, Namibia, Botswana, and Tanzania; in all other diamond-producing countries such as Angola, the Central African Republic, Guinea, and Sierra Leone alluvial mining dominated (US GAO, 2002); further details on the diamond industry are provided in Appendix Section A. As diamonds are small, easily concealed, and extremely valuable, the risk of theft or smuggling is high. Without tamper-proof packaging and a certification system in place, it is difficult to trace the origin of diamonds, especially when diamonds from different sources are mixed (US GAO, 2002).

In the late 1990s, civil society groups and the United Nations increasingly called attention to the role of so-called "blood" or "conflict" diamonds in various brutal wars, such as in the Democratic Republic of the Congo, Angola, Liberia, and Sierra Leone (e.g. Haufler, 2009; Bieri, 2015). The NGOs Global Witness and Partnership for Africa published reports docu-

2018; Bloem, 2023).

menting how the illicit sale of rough diamonds had fueled armed conflict in Angola and Sierra Leone, respectively (Global Witness, 1998; Smillie et al., 2000). As a response, in 1998, the UN Security Council banned the trade in diamonds from Angola, unless a certificate confirmed the diamonds originated from areas under the control of the government. Similar sanctions were imposed two years later on Sierra Leone (Wright, 2004). However, these sanctions were not effective, as rebel groups were able to smuggle rough diamonds via neighboring countries, where certificates of origin were not required (Global Witness, 1998; US GAO, 2002).

Continued global concern over "blood diamonds" prompted various stakeholders to take action. In May 2000, South Africa, Botswana, and Namibia convened the first meeting in Kimberley, South Africa, marking the beginning of the Kimberley Process. The Kimberley Process involved three stakeholders: governments involved in the production, consumption, and trade of diamonds; NGOs; and industry representatives who were organized under the aegis of the newly founded World Diamond Council (Wright, 2004; Haufler, 2009; Bieri, 2015). The establishment of a certification scheme was backed by the United Nations General Assembly in its Resolution A/RES/55/56 in December 2000. Participants in the Kimberley Process were requested to report on their progress at the next session of the General Assembly. In its Resolution A/RES/56/263, adopted in March 2002, the UN General Assembly urged participants to finalize the certification scheme and to immediately commence with its implementation. Participants in the Kimberley Process met in the same month. A document from the United States General Accounting Office from June 2002 reports that the remaining technical issues regarding implementation were resolved at this meeting. To move ahead with implementation, there was agreement to focus on adopting the scheme at the national level. Furthermore, and importantly, the report confirms that the involved countries pushed for rapid implementation: "Those in a position to issue the Kimberley Process Certificate were asked to do so immediately. All others were encouraged to do so by June 1, 2002" (US GAO, 2002, p. 7). At the next meeting in November 2002, the Kimberley Process Certification Scheme was formally adopted and January 2003 was set as the official starting date.

As Haufler (2009, p. 4) writes, "The Kimberley Process is an industry-based certification scheme wrapped inside an export/import regime that is implemented through domestic legislation in member states. It is designed to track rough diamonds and prevent those from conflict zones from entering legitimate world markets." The KPCS aims to ensure that members trade only with other members. While the scheme as such is voluntary, it creates strong incentives for countries (and firms) to be part of the "club" (Haufler, 2009; Bieri, 2015).[7] Sixteen African countries participated in the KP from the very start: Angola, Botswana, the Central African Republic, the Democratic Republic of the Congo, Ghana, Guinea, Ivory Coast, Lesotho, Mauritius, Namibia, the Republic of Congo, Sierra Leone, South Africa, Tanzania, Togo, and Zimbabwe (Appendix Tables A2 and A3). This list includes all major diamond-exporting countries (see Appendix Section A).

[7] See also the official website of the KPCS (link).

As mentioned, the KPCS is not an international treaty or agreement. Rather, governments are obligated to implement national legislation in compliance with KPCS standards. This has the advantage of making the KPCS standards legally binding in member states, rather than just a recommendation that industry players are expected to follow. At the same time, however, the KPCS relies on the willingness and ability of member states to monitor and enforce the scheme. This has, therefore, raised concerns that the scheme lacks sufficient rigor, particularly in countries with weak governance and high levels of corruption (Smillie, 2005; Haufler, 2009; Smillie, 2014; Bieri, 2015). Additionally, the KPCS lacks an institutional setup. For example, it has no headquarters, no staff, and no budget (Bieri, 2015). There are limited possibilities to discipline participants, and hardly any members have been expelled to date (Smillie, 2005; Haufler, 2009).

In practice, as Engwicht (2018) highlights for the case of Sierra Leone, the actual enforcement of the scheme may be less crucial to the success of the KPCS, provided governments are able and willing to ensure – via formal or informal means – that diamonds entering the formal market do not finance rebel groups. If most diamond-producing countries are successful in this endeavor, rebel groups will face significant challenges in bringing diamonds into the world market.

3 Main Data

Following recent work (e.g. Michalopoulos and Papaioannou, 2016; Berman et al., 2017), we conduct the analysis at the 0.5 × 0.5 degree grid cell level (approximately 55 km × 55 km at the equator). This allows us to circumvent the use of administrative units as the unit of observation, which may be endogenous to conflict (Berman et al., 2017; Rigterink, 2020). For our analysis, we draw on the PRIO-GRID cell data include a total of 10,675 grid cells for the African continent (Tollefsen et al., 2012). We remove grid cells, mostly coastal and island grid cells, with a land area in the lowest 5th percentile. We also remove grid cells that have missing values for control variables (most of these cells are in the lowest 5th percentile for land area). Our sample thus includes a total of 10,125 grid cells.

3.1 Study Period

In most analyses, we restrict the study period to the years 1997 to 2013. We start in 1997 because this is when ACLED data first became available, while we end in 2013 in light of the rise in violent conflicts in the 2010s, which seems to have disproportionately affected grid cells without SD suitability (see Figure 2a).[8] That is, for most analyses, we consider the first 12 years post-KPCS. When examining pre-trends and the long-run impact of the KPCS, we consider all years for which we have data about armed conflicts (i.e. from 1997 onwards in the case of

[8] This aligns with McGuirk and Nunn (2025), who show that reduced rainfall in transhumant pastoralist areas has driven conflicts with nearby sedentary agriculturalists due to pre-harvest migrations.

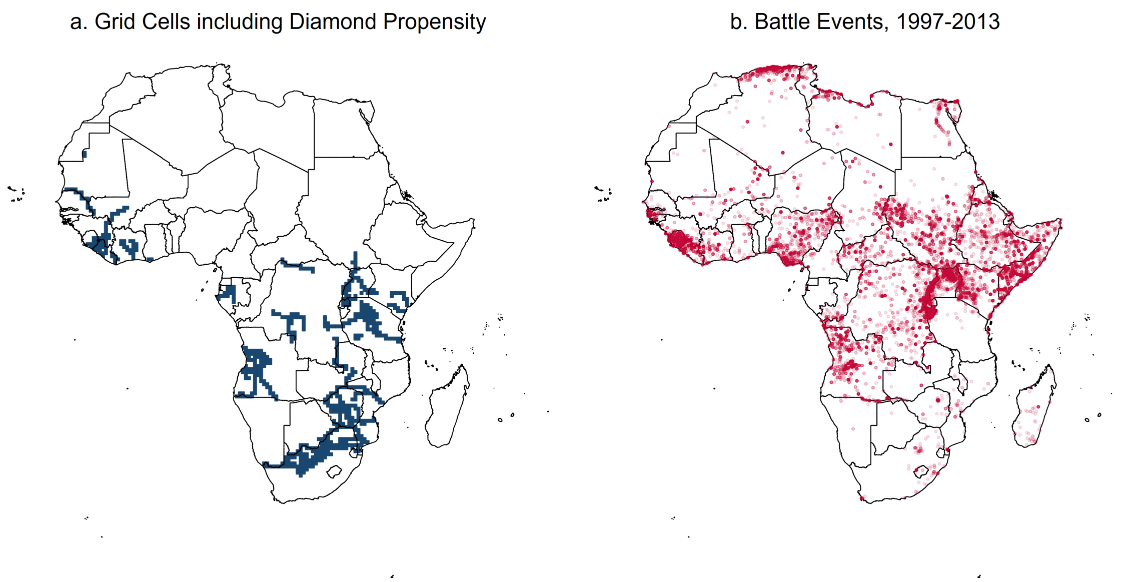
ACLED and from 1989 onwards in the case of UCDP). Note that 2020 is the last year included, as the time-varying controls are available only up to then.

3.2 Measure for Secondary Diamond Suitability

We use data on secondary diamond *suitability* from Rigterink (2020) rather than data on deposits. There are two reasons for this. First, artisanal and small-scale mining activities are likely endogenous to armed conflict. Second, data on small-scale mining activities are often inaccurate due to difficulties measuring such activities (Rigterink, 2020). Determining secondary diamond suitability follows a two-step procedure. The first step determines locations that are suitable for primary diamonds. These are locations with kimberlite or lamproite deposits that intersect an archon, an area of a particular geological age. The second step determines whether a grid cell is suitable for secondary diamonds by exploiting that secondary diamonds are typically found along or near downstream river segments (up to 600km) from a host rock. Drawing on available information on the world's rivers and their flow direction, the procedure identifies rivers located within a 750m radius of a potential host rock, and subsequently traces these rivers downstream up to a radius of 600km from the potential host rock, determining the total length in kilometers of relevant river segments in a grid cell.[9] We create a binary variable that equals 1 for grid cells with any such river segment (irrespective of its length), and 0 otherwise. Out of all 10,125 grid cells, 178 (less than 2%) constitute grid cells with a suitability for primary diamonds, and 689 (6.8%) constitute grid cells with a suitability for secondary or alluvial diamonds. Panel (a) of Figure 1 shows the spatial distribution of grid cells with secondary diamond suitability. The black lines indicate the country's borders.

To show that secondary diamond suitability is predictive of actual diamond occurrence, we correlate our binary suitability measure with grid cells containing verified diamond deposits. Based on DIADATA (Gilmore et al., 2005), there are only 212 (2.09%) grid cells with diamond occurrence, compared to 689 (6.8%) grid cells with secondary diamond suitability. This is not surprising, however, given that data on small-scale mining activities are often inaccurate, as mentioned above. Using these data, Table 1 shows that the suitability and the occurrence variables are strongly correlated. This holds when we include a set of controls — total annual precipitation and average annual temperature (linear and squared terms), latitude, longitude, elevation, ruggedness, and distance to the coast (for details on these variables, see below) — and region or country fixed effects. In the case of country fixed effects, we obtain an estimate of 0.088 for secondary diamond suitability (p-value: 0.024, column 4), i.e., secondary diamond suitability grid cells are 8.8 pp more likely to have diamond occurrence. This suggests a strong relationship between SD suitability and occurrence, considering that the sample mean for diamond occurrence is equal to 2.1%.

[9] The 750-meter threshold is based on the largest known kimberlite deposit. If no such river exists, the closest river to a potential host rock is selected. These rivers lie within 5km of a potential host rock.



Notes: The figure displays in Panel (a) the spatial distribution of grid cells with secondary diamond suitability and in Panel (b) the spatial distribution of all battle events reported by ACLED over the period 1997–2013 (georeferenced). The black lines indicate country borders.

Figure 1: Spatial Pattern of Secondary Diamond suitability and Armed Conflicts

Table 1: The Association between Secondary Diamond Suitability and Occurrence

	Secondary Diamond Occurrence			
	(1)	(2)	(3)	(4)
Secondary Diamond Suitability	0.113***	0.098***	0.101***	0.088***
	(0.033)	(0.028)	(0.028)	(0.024)
Observations	10,125	10,125	10,125	10,125
R^2	0.040	0.063	0.065	0.102
Controls	No	Yes	Yes	Yes
Region FE	No	No	Yes	No
Country FE	No	No	No	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports estimates from regressing a dummy for secondary diamond occurrence (from Gilmore et al., 2005) on a dummy for secondary diamond suitability. Controls include total annual precipitation and average annual temperature (linear and squared terms) as well as latitude, longitude, elevation, ruggedness, and distance to the coast. Region refers to the main geographic regions (North, East, South, West, Middle) as defined in Christensen et al. (2024). We compute Conley standard errors allowing for spatial correlation within a radius of 500 km.

3.3 Armed Conflict Data: ACLED and UCDP

We draw on the two widely used conflict datasets, the Armed Conflict Location & Event Data Project (ACLED) (Raleigh et al., 2010) and the UCDP/PRIO Armed Conflict Dataset, Version 22.1 (Gleditsch et al., 2002; Davies et al., 2022). ACLED classifies violent events into three subcategories: battles, explosions/remote violence, and violence against civilians. According to the official KP website, "conflict" or "blood" diamonds "are rough diamonds used by rebel movements or their allies to finance armed conflicts aimed at undermining legitimate governments" (link). Similarly, the United Nations General Assembly understands "conflict diamonds to be rough diamonds which are used by rebel movements to finance their military activities, including attempts to undermine or overthrow legitimate governments."[10] We therefore focus in our analysis on "battles," which are defined as "a violent interaction between two politically organized armed groups at a particular time and location" (ACLED, 2023, p. 8). Appendix Figure A1 displays the total number of armed conflicts by year, as well as the share of grid cells with at least one armed conflict. It documents a decline in armed conflict from the late 1990s to the mid-2000s, followed by an increase in armed conflict, particularly after 2010. Conflict events are geo-referenced, allowing us to assign them to specific grid cells. Our main dependent variable is a dummy variable that equals 1 for grid cells in which at least one battle occurred in a given year. Based on this variable, a total of 6,541 conflict events occurred between 1997 and 2013. Panel (b) of Figure 1 shows the spatial distribution of all battle events reported by ACLED over the period 1997–2013.

UCDP collects data on state-based armed conflicts, non-state conflicts, and one-sided violence. In contrast to ACLED, it only considers armed conflicts that resulted in at least 25 combat-related deaths in at least one calendar year in which the conflict took place. Based on UCDP, there were a total of 5,104 such conflict events during our study period (compared to 6,541 based on ACLED.) The pattern over time largely mirrors the pattern based on ACLED (see Appendix Figure A1). A discrepancy occurs after 2010, when ACLED reports a sharper increase in armed conflicts compared to UCDP.

3.4 Other Data

We obtain annual grid cell level data on total precipitation and average temperature from Afro-Grid (Schon and Koren, 2022), which covers the period from 1989 to 2020. We determine the distance of a grid cell to the coast based on its geographic center (centroid) and define the distance variable as the log of one plus the distance from a grid cell's centroid to the coast. Grid cells that include a coast are defined as having a distance of 0. We use elevation data at the 15 arc-second level based on SRTM X-SAR Digital Elevation Models created by the German Aerospace Center (DLR), the Italian Space Agency (ASI), and NASA/JPL and corrected by Jonathan de Ferranti (link). Elevation and ruggedness are defined as the average elevation and the standard deviation of elevation within each grid cell, respectively. Finally, we use data on a grid cell's suitability for alluvial gold mining from Girard et al. (2025). For a location to be considered suitable for artisanal and small-scale gold mining (ASM), the bedrock must satisfy the following conditions: (i) It comes from a geological age linked to gold mineralization, and (ii) it has a rock type (lithology) that can host gold.

[10] United Nations General Assembly Resolution 55/56 (January 29, 2001).

4 Empirical Approach

To estimate the causal impact of the KPCS on the incidence of armed conflict in Africa, we employ a difference-in-differences strategy. We compare over time the incidence of armed conflict between grid cells with and without secondary diamond (SD) suitability, controlling for grid cell and year fixed effects as well as a set of time-varying controls. The analysis uses data from 1997 for ACLED and from 1989 for UCDP up to the year 2020. Because participants in the Kimberley Process were asked to begin implementing the certification scheme in 2002 (compare Section 2), we consider 2002 as the first post-treatment year. More formally, we estimate the following model:

$$Y_{it} = \alpha + \sum_{t=1997,1989}^{2020} \beta_t \left(SecondaryDiamondSuitability_i \times Year_t \right) + \delta_i + \phi_{rt} + \Gamma X_{it} + \sum_{t=1997,1989}^{2020} \Phi \left(C_i \times Year_t \right) + \epsilon_{it}, \quad (1)$$

where $Y_{i,t}$ is equal to 1 if ACLED (UCDP) reports at least one armed conflict in grid cell i and year t, and zero otherwise. $SecondaryDiamondSuitability$ is an indicator variable and is equal to 1 for grid cells with a secondary diamond suitability greater than zero. We interact $SecondaryDiamondSuitability$ with the year dummies ($Year_t$), with the omitted year being 2001. We include grid cell fixed effects (δ_i) to control for any time-constant grid cell level characteristics such as geographic characteristics and, following recent work by Christensen et al. (2024), we include region $\times$ year fixed effects (ϕ_{rt}) to control for any changes in conflict over time that are common across grid cells within the same region.[11]

Past work has shown that climatic conditions may fuel violent conflict (e.g. Harari and Ferrara, 2018; Eberle et al., 2025; Fetzer, 2020). To address the concern that the estimated β_ts may partially capture changes in armed conflict that are due to other changes over time, we additionally include a set of time-varying controls. These are time-varying variables (X_{it}) – temperature, temperature squared, precipitation, precipitation squared – and interactions between time-invariant variables (C_i) – latitude, longitude, elevation, ruggedness, and the log of (1 + a grid cell's distance to the coast) – and the year dummies. We follow Berman et al. (2017) and compute Conley (1999) standard errors, allowing for spatial correlation within a radius of 500 km (which is close to the median internal distance of African countries) and infinite serial correlation. Our estimates are robust to using other radii between 100 and 2,000

[11] Christensen et al. (2024) define five main regions, East, Middle, North, South, and West, see Appendix Figure A2. In Appendix Table A5, we show that the estimated impact of the KPCS on armed conflict is robust to the inclusion of country $\times$ year fixed effects. The estimates are smaller in absolute terms, as expected, because of the geographic spillovers we document in Section 6.1. Grid cells neighboring SD suitability grid cells also experience a reduction in armed conflict after the introduction of the KPCS. Nonetheless, the estimates remain economically meaningful and statistically significant at the 1% level (column 1 of Appendix Table A5).

km (see Section 5.2).

We use the event-study specification to test for parallel trends in armed conflict between SD suitability and non-SD suitability grid cells prior to the introduction of the KPCS. Specifically, we evaluate the significance of the $\beta_{t\leq 2000}$ coefficients and, following Roth (2022), compute the power to detect a linear violation of parallel trends (see Section 5.1). In addition, we use the following simple difference-in-differences model to estimate the overall impact of the introduction of the KPCS on armed conflicts in the first 12 years:

$$Y_{it} = \alpha + \beta \left(SecondaryDiamondSuitability_i \times Post_t \right) + \delta_i + \phi_{rt} + \Gamma X_{it} + \sum_{t=1997}^{2013} \Phi \left(C_i \times Year_t \right) + \epsilon_{it}, \quad (2)$$

where $Post_t$ is equal to 1 for all years following the introduction of the KPCS (2002-2013); for the years 1997–2001 it is zero (see Section 3.1 for an explanation of the study period). All other variables are defined as before.

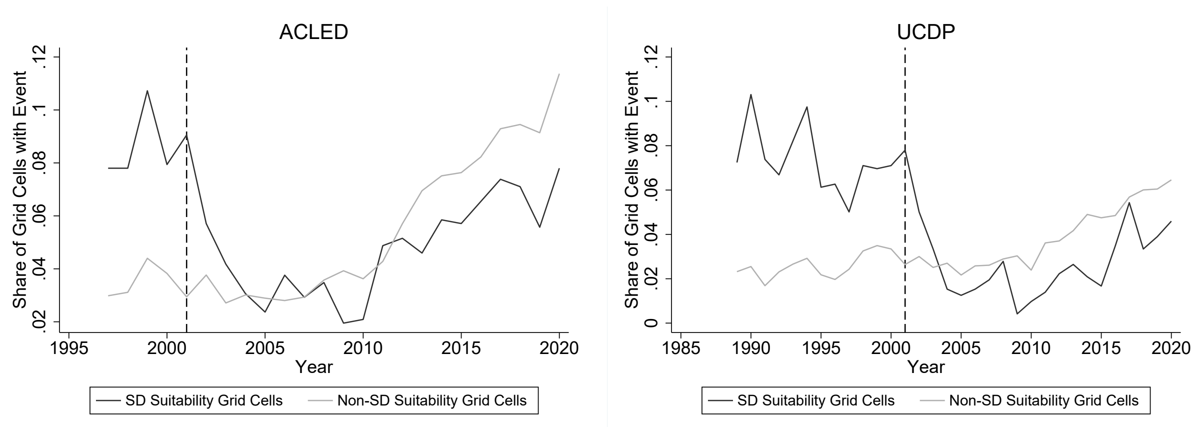
Our identification strategy requires that, conditional on grid cell and region × year fixed effects as well as a set of time-varying controls, a grid cell's suitability for secondary diamonds is not correlated with other factors that may have reduced armed conflict over time. In addition to testing for parallel trends using equation 1, we address different concerns about our identification strategy and the robustness of our results in Section 5.2. Furthermore, we provide evidence that supports our claim that the certification scheme indeed causes the observed effects. Exploiting variation in treatment intensity, we show in Section 5.5 that the effects are comparatively larger for SD suitability grid cells in countries that participated in the Kimberley Process from its start and for SD suitability grid cells with above median length of river segments into which diamonds may have been carried. Finally, we use a rebel group-level analysis to confirm that the KPCS reduced the armed activities of rebel groups and that their activities have become more geographically confined (Section 6.2).

5 The Impact of the KPCS on Armed Conflicts

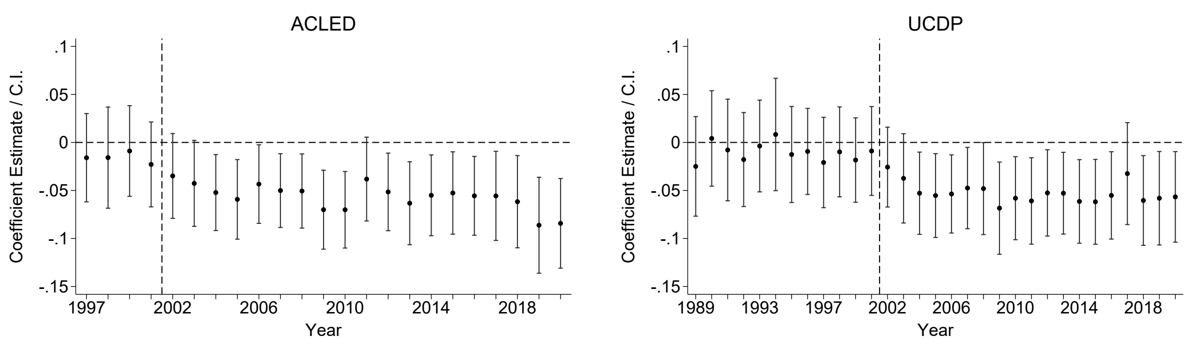
5.1 Main Results

Figure 2a shows, by year, the share of grid cells with and without secondary diamond (SD) propensity for which ACLED (left panel) or UCDP (right panel) report at least one armed conflict. The figure documents two facts. First, prior to 2002, SD suitability grid cells saw on average more armed conflict compared with non-SD suitability grid cells (8.7% compared to 3.5%; see Appendix Table A4). Second, following the introduction of the KPCS in 2002, the incidence of armed conflict declined sharply for SD suitability grid cells while it remained flat for non-SD suitability grid cells.

(a) Share of Grid Cells with Armed Conflicts over Time



(b) Difference-in-Differences Estimates



Notes: Figure 2a shows, by year, the share of grid cells with and without secondary diamond propensity for which ACLED (left panel) and UCDP (right panel) reported at least one armed conflict. FIgure 2b plots the difference-in-differences estimates based on ACLED (left panel) and UCDP (right panel) of the interactions between a dummy for grid cells with secondary diamond suitability and year fixed effects with their 95% confidence intervals. The omitted year is 2001. The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. Regressions include grid cell and region $\times$ year fixed effects as well as a set of time-varying controls, including (i) temperature, temperature squared, precipitation, precipitation squared, and (ii) interactions between time-invariant controls (latitude, longitude, elevation, ruggedness, and the log of 1 + a grid cell's distance to the coast) and the post-KPCS dummy. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation. The dashed vertical line indicates the start of the KPCS in 2002.

Figure 2: The Effect of the Kimberley Process on Armed Conflicts

Figure 2b plots the event-study difference-in-differences estimates based on equation (1). The figure documents the existence of parallel trends prior to the introduction of the KPCS when controlling for grid cell and region $\times$ year fixed effects and a set of time-varying controls. When using UCDP, which allows us to examine trends from 1989 onwards, the break between the pre-KPCS and the post-KPCS years is even more pronounced (right panel of Figure 2b). For both data sets, the estimates are small and not statistically different from zero for each time period prior to the KPCS. Moreover, we cannot reject the null hypothesis that the $\beta_{t<2001}$ in equation (1) are jointly zero.[12] Importantly, the tests are sufficiently powered to detect meaningful linear violations of parallel trends.[13] Together, this supports our identifying assumption.

Column (1) of Table 2 reports the results from estimating model (2). Panel (a) presents the results for ACLED and Panel (b) for UCDP. We observe the same absolute estimate in both data sets. The estimates of -0.041 imply an economically large reduction in armed conflict over the 2002–2013 period by 47% and 59% for ACLED and UCDP, respectively. This is based on the mean armed conflict incidence for SD suitability grid cells over the 1997-–2001 period, which we also report in the table. Both estimates are significant at the 1% level.

5.2 Robustness

In the following, we provide further supportive evidence for the validity of our identification strategy and the robustness of our results.

Removing High-Conflict Observations. We first examine the extent to which potentially high-conflict observations drive the observed effect. For example, the wars in Angola and Sierra Leone ended in 2002. At the same time, these are countries with substantial alluvial mining, and diamonds are considered to have played an important role in both wars (Global Witness, 1998; Smillie et al., 2000; US GAO, 2002). We therefore first remove grid cells located in Angola and Sierra Leone from the sample. Doing so, we obtain an estimate of 0.020 for both ACLED and UCDP (column 2 of Table 2). Since the mean armed conflict incidence pre-KPCS for SD suitability grid cells outside Angola and Sierra Leone is 0.054 (vs. 0.087 for the full sample), the relative effect size is still large (37% vs. 47%).

As other countries also saw an end to violence following the introduction of the KPCS, and wars in one country may impact the conflict dynamics in neighboring countries, we next conduct the following two exercises. First, we re-estimate the baseline specification, dropping one country at a time and, alternatively, two countries at a time. Figure A3 plots the distribution of the difference-in-difference estimate and shows that the distribution is centered around our main estimate of -0.041 and that the distribution peaks slightly to the left of -0.041 (for both ACLED and UCDP). In addition, the estimates we obtain when removing Angola and Sierra Leone are in the very right tail of the distributions. Second, we examine the robustness

[12] The Wald test statistic (p-value) for the joint hypothesis test of zero pre-KPCS effects is 1.183 (0.881) for ACLED and 4.363 (0.976) for UCDP.

[13] Following Roth (2022), we compute the power of detecting a linear violation of the parallel trends assumption using our event-study estimates. For the ACLED data, we have four pre-treatment periods (left panel of Figure 2b). The slope of a linear pre-trend that we would detect half of the time is -0.01. Power rises to 80% if we allow a linear pre-trend with a slope of -0.015, which would capture the trend of the estimated conflict reduction in the first four post-treatment years. Having more pre-treatment periods (UCDP data, right panel of Figure 2b), allows us to test even less severe pre-trend violations. For 50% power, the slope for any significant pre-treatment coefficient is about -0.003, and for 80% power, we would detect a pre-trend violation with a slope of -0.005. Overall, we consider it unlikely that any violation of parallel trends significant enough to explain our results would have gone undetected.

Table 2: The Effect of the Kimberley Process on Armed Conflicts: Difference-in-Differences Estimates

	Incidence of Armed Conflict						
	Baseline					River	Gold
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: ACLED							
SD Suitability × Post	-0.041***	-0.020***	-0.031***	-0.031***	-0.030***	-0.015*	-0.041**
	(0.009)	(0.007)	(0.009)	(0.009)	(0.012)	(0.009)	(0.010)
Mean Incidence pre-KPCS	0.087	0.054	0.082	0.087	0.087	0.094	0.072
Observations	172,125	164,662	169,099	172,125	172,125	60,571	74,885
R^2	0.319	0.326	0.319	0.319	0.319	0.330	0.322
Panel B: UCDP							
SD Suitability × Post	-0.041***	-0.020***	-0.040***	-0.034***	-0.031**	-0.023***	-0.045***
	(0.009)	(0.007)	(0.010)	(0.009)	(0.012)	(0.009)	(0.011)
Mean Incidence pre-KPCS	0.070	0.038	0.072	0.070	0.070	0.077	0.058
Observations	172,125	164,662	169,099	172,125	172,125	60,571	74,885
R^2	0.323	0.330	0.323	0.324	0.323	0.300	0.301
Grid-Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time-Varying Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
W/o Angola & Sierra Leone	No	Yes	No	No	No	No	No
W/o PD Suitability Cells	No	No	Yes	No	No	No	No
Dist. Ind. Mine × Year FE	No	No	No	Yes	No	No	No
SD Suit. × Mineral Prices	No	No	No	No	Yes	No	No

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Difference-in-differences estimates on the interaction between a dummy for grid cells with secondary diamond (SD) suitability and the post-KPCS dummy. The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. All regressions include grid cell and region × year fixed effects as well as a set of time-varying controls. These controls include (i) temperature, temperature squared, precipitation, precipitation squared, and (ii) interactions between time-invariant controls (latitude, longitude, elevation, ruggedness, and the log of (1 + a grid cell's distance to the coast)) and the post-KPCS dummy. The study period is from 1997-2013 and countries started to implement the KPCS in 2002. 'Mean incidence pre-KPCS' gives the mean armed conflict incidence for secondary diamond suitability grid cells over the 1997–2001 period. Column (1) reports our baseline specification. Column (2) removes grid cells located in Angola and Sierra Leone. Column (3) removes grid cells with primary diamond (PD) suitability. Column (4) adds interactions between the log of (1 + a grid cell's distance to an industrial diamond mine (pre-KPCS)) and year fixed effects. Column (5) adds interactions between the dummy for secondary diamond suitability and the world prices of five important minerals in Africa (diamonds, gold, silver, copper, and tin). Column (6) restricts the sample to grid cells with alluvial gold suitability. Column (7) restricts the sample to grid cells that have a major river within 25km from their border. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation.

of our estimates when we remove the grid cells of one of the five regions from the sample at a time. The results are shown in Appendix Table A6. For ACLED, the coefficient estimates range from -0.029 when dropping the central region to -0.049 when dropping the western or eastern region; all estimates are significant at the 1% level. As the mean incidence of armed conflict pre-KPCS for secondary diamond suitability grid cells is lowest when we remove the central region (0.058 vs. 0.087), the relative effect size remains similar. The estimates based on UCDP provide a similar pattern.

Increased Regulation of Industrial Diamond Mining. Over the considered time period, several transparency schemes and national regulatory acts were initiated to regulate the mining sector (Berman et al., 2017; Christensen et al., 2024). Two such measures were launched around the same time as the KPCS. The International Council on Mining and Metals (ICMM, https://www.icmm.com/), which promotes CSR reporting, was created in 2001. Furthermore, beginning in 2004 the US started to enforce the Foreign Corrupt Practices Act (FCPA). One concern is therefore that our estimates may be distorted by the (potential) impact these initiatives had on armed conflict. To address this concern, we use the fact that both measures target industrial rather than alluvial mining. Thus, we drop grid cells with a suitability for primary diamonds. This is feasible because primary diamonds can be extracted at the host rock only, while alluvial diamonds can also be extracted up to 600 km away (see Section 3). While we obtain somewhat smaller estimates compared to those based on our baseline specification, the estimated effect remains large at 3.1 percentage points; see column (2) of Table 2.

Rise in World Mineral Prices. Berman et al. (2017) show that over the 1997–2010 period, the historical rise in world mineral prices (commodity super-cycle) increased violence in Africa in grid cells with a (large-scale) mine. To rule out that part of the decline in violent conflict we observe is related to changes in mineral prices and/or (other) changes in industrial mining activities, we run the following two exercises. First, we add interaction terms between the log of one plus a grid cell's distance from its centroid to the closest industrial diamond mine pre-KPCS and year fixed effects.[14] Second, we add interaction terms between the dummy for secondary diamond suitability and the world price of each of the five important minerals in Africa: diamonds, gold, silver, copper, and tin. The world price data for minerals and data on the location of industrial diamond mines prior to the introduction of the KPCS (in 2001) come from Berman et al. (2017). They highlight that including diamond prices is problematic because they depend on the quality and type of diamonds (data that is not available), and prices for different qualities may not move in the same direction. They therefore exclude diamond prices in their baseline specification. The results of the two exercises are shown in columns (3) and (4) of Table 2. The coefficient estimates are somewhat smaller (between -0.030 and -0.036) compared to those from the baseline specification (column 1), but remain statistically significant.

Increasing the Comparability Between Treatment and Control Group. In the last two columns of Table 2, we restrict the sample to a certain subset of grid cells to increase the comparability between the treatment and the control group (SD and non-SD suitability grid cells). In column (5), we aim to restrict the sample to grid cells with a river. As available

[14] We obtain similar results when we add instead an interaction term between the log of (1+a grid cell's distance from its centroid to the closest industrial diamond mine pre-KPCS and the world diamond price (P_t^D).

data from naturalearthdata.com provides information only for major rivers, we consider a grid cell as (likely) having a river if there is a major river within 25km from its border. This leaves us with only 3,563 grid cells (out of 10,125), of which 484 (or 13.6%) are suitable for secondary diamonds. When restricting the sample to these grid cells, the estimates are smaller. For ACLED, the relative effect size is 21% (vs. 47% in column 1); for UCDP, the relative effect size is 32% (vs. 59% in column 1). Yet, they remain statistically significant.

In column (6), we restrict the sample to grid cells with certain geographic features, namely grid cells with strictly positive alluvial gold suitability, drawing on new grid cell level data from Girard et al. (2025). There are 4,405 grid cells with alluvial gold suitability, with 485 grid cells suitable for both alluvial gold and diamond mining. Reassuringly, we obtain similar estimates (and relative effects) to the baseline specification (column 1) when restricting the sample to gold-suitability grid cells.

Spatial Correlation. In our baseline specification, we compute Conley (1999) standard errors, allowing for spatial correlation within a radius of 500 km and infinite serial correlation. As a final exercise, we examine how sensitive our results are to the use of radii ranging from 100 to 2,000 km in steps of 100 km, and infinite serial correlation. The results are shown in Appendix Figure A4. It illustrates that most of these radii yield even (slightly) smaller standard errors than when using a radius of 500 km.

5.3 Alternative Measures for Armed Conflict and Other Forms of Violence

Until now, we have focused on the incidence of armed conflicts as our outcome variable. Using ACLED, we will now consider alternative outcomes, including conflict intensity. In addition, we examine whether rebel groups resorted to other forms of violence in response to increased difficulties in generating revenue. The results are reported in Table 3. The event-study difference-in-differences estimates are plotted in Appendix Figure A5.

In column (1), we first consider only battle events with fatalities. Out of 5,615 battle events over the 1997–2011 period, 37.9% involved no fatalities. Accordingly, the pre-KPCS mean in conflict incidence for SD suitability cells is now only 4.2% (instead of 8.7%). We obtain a difference-in-difference estimate of -0.022, which implies a reduction of armed conflict by 52% (compared to 47% based on our baseline specification, column 1 of Table 2).

In columns (2) and (3), we consider two measures of conflict intensity: the number of battles in grid cell i and year t and the number of fatalities in grid cell i and year t. We take the natural log of one plus the number of battles/fatalities due to the highly skewed distribution of both variables.[15] While 96.2% of grid cell-years experienced no battles, those with the most intense fighting saw over 50 battles. Similarly, 97.5% of grid cell-years had no fatalities, whereas others had several hundred or several thousand fatalities. In column (4), we therefore top-code

[15] We obtain similar results when using the inverse hyperbolic sine (IHS) transformation of the total number of battles/fatalities.

Table 3: The Effect of the Kimberley Process on Armed Conflicts Using Alternative Measures and on Other Forms of Violence

	Armed Conflict				*Other Violence*	
	Any Event	Ln(1+..)			Any Event	
	Battles w/ Fatalities	Nr Battles	Nr Fatalities	Nr Fatalities Top-Coded	Violence Against Civilians	Any Violent Event
	(1)	(2)	(3)	(4)	(5)	(6)
SD Suitability × Post	-0.022***	-0.075***	-0.087***	-0.057***	-0.026***	-0.043***
	(0.006)	(0.019)	(0.024)	(0.015)	(0.009)	(0.011)
Mean Incidence pre-KPCS	0.042	0.704	32.1	0.442	0.084	0.131
Observations	172,125	172,125	172,125	172,125	172,125	172,125
R^2	0.015	0.024	0.013	0.014	0.018	0.026
Grid-Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Time-Varying Controls	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Difference-in-differences estimates on the interaction between a dummy for grid cells with secondary diamond (SD) suitability and the post-KPCS dummy. All regressions include grid cell and region × year fixed effects as well as a set of time-varying controls. These controls include (i) temperature, temperature squared, precipitation, precipitation squared, and (ii) interactions between time-invariant controls (latitude, longitude, elevation, ruggedness, and the log of 1 (+ a grid cell's distance to the coast)) and the post-KPCS dummy. The study period is from 1997-2013 and countries started to implement the KPCS in 2002. 'Mean incidence pre-KPCS' gives the mean armed conflict incidence for secondary diamond suitability grid cells over the 1997–2001 period. In columns 2-4, it gives the mean value for the non-transformed variable. In column (1), the dependent variable is an indicator variable equal to 1 if at least one armed conflict with fatalities occurred in grid cell i and year t, and 0 otherwise. In column (2), the dependent variable is the natural log of (1+ the number of battle events in grid cell i and year t). In column (3), it is the natural log of (1+ the number of fatalities in grid cell i and year t) while in column (4) the dependent variable is the same as in column (3), but top-coded at the 99th percentile (14 fatalities). In column (5), the dependent variable is an indicator variable equal to 1 if at least one violent event against civilians occurred in grid cell i and year t, and 0 otherwise. In column (6), the it is an indicator variable equal to 1 if any violent event occurred in grid cell i and year t, and 0 otherwise. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation.

the fatality variable at the 99th percentile (14 fatalities). For all three outcomes, the estimate on the interaction term between SD suitability and the post dummy is negative and significant at the 1% level.

One concern is that by focusing on battles, we may have missed important changes in the type of violence that rebel groups engage in. In column (5), we therefore consider violence against civilians (the second largest category of violent events in which rebel groups are involved), and in column (6), we consider any violent event; this category includes, in addition to battles and violence against civilians, explosions/remote violence. We can thereby rule out the possibility that the documented decline in armed conflict came at the cost of other forms of violent conflict. Over the 1997–2001 period, 8.4% (13.1%) of SD suitability grid cells experienced at least one event coded as violence against civilians (any violent event). The estimate in column (5) indicates a decrease in violence against civilians, albeit of smaller magnitude. When we consider all violent events (column 6), we obtain an estimate of -0.043, which implies a relative decline by 33%.

Overall, these results support our main finding of a significant reduction in armed conflict and rule out an increase in violent conflict other than battles following the introduction of the KPCS.

5.4 Placebo Test

Another way to stress test our identification strategy is to use the introduction of the KPCS as a placebo event. The KPCS aims to limit trade in certified alluvial diamonds to prevent rebel groups from financing their military activities through the sale of rough diamonds. Therefore, it should not alter the incidence of armed conflict related to non-diamond artisanal and small-scale mining. We focus on artisanal gold mining using new grid cell level data on alluvial gold mining suitability from Girard et al. (2025), as this is the most common form of alluvial mining in Africa. As mentioned above, there are a total of 4,405 cells with a suitability for alluvial gold mining; 485 cells are suitable for both alluvial gold and diamond mining.

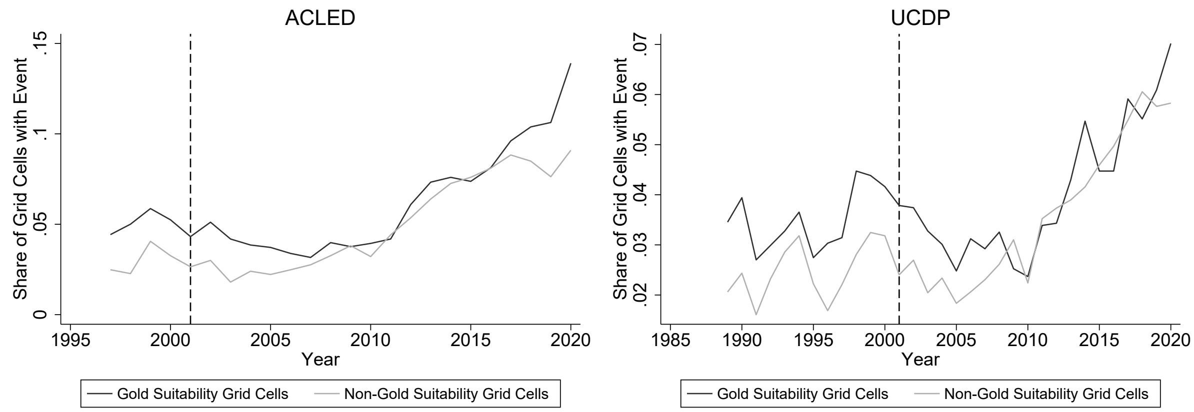
Figure 3a shows descriptively that after the KPCS was introduced, the trend in the incidence of armed conflict remained similar for grid cells with and without gold suitability. As a more formal test, we reestimate our baseline specification, replacing the interactions between the period dummies and a dummy for SD suitability grid cells with interactions between the period dummies and a dummy for alluvial gold suitability grid cells. Figure 3b plots the estimates of these interaction terms. It shows that there is no differential trend in the incidence of armed conflict for gold suitability cells prior to the KPCS as well as following its introduction in 2002. This indicates that the impact of the KPCS is specific to alluvial diamond mining.

5.5 Heterogeneity by Treatment Intensity

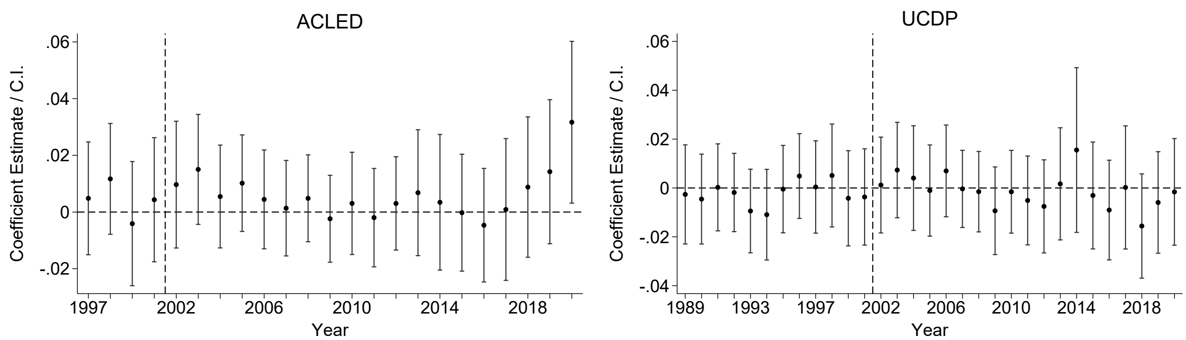
This section presents supportive evidence that the differential effect observed after 2001 for grid cells with secondary diamond suitability represents the causal effect of the KPCS on armed conflict in Africa. We do so by documenting that the impact of the KPCS on armed conflict is greater for grid cells that were subject to greater treatment – specifically, grid cells located in countries that were part of the KPCS from the outset and grid cells with a high suitability for alluvial diamond mining.

Early KP vs. Late/Non-KP Member Countries. We first test the hypothesis that the observed decline in armed conflict is larger for secondary diamond suitability grid cells located in countries that were part of the Kimberley Process from its start. Of the 24 countries with secondary diamond suitability grid cells (out of 54 African countries), 12 countries joined directly: Angola, Botswana, CAR, DRC, Côte d'Ivoire, Ghana, Guinea, Namibia, Sierra Leone, South Africa, Tanzania, Zimbabwe (with the remaining countries being Gabon, Kenya, Liberia, Mali, Mauritania, Mozambique, Rwanda, Senegal, Sudan, Swaziland, Uganda, and Zambia). To

(a) Share of Grid Cells with Armed Conflicts over Time



(b) Difference-in-Differences Estimates



Notes: Figure 3a shows, by year, the share of grid cells with and without alluvial gold suitability for which ACLED (left panel) and UCDP (right panel) reported at least one battle event. Figure 3b plots the difference-in-differences estimates based on ACLED (left panel) and UCDP (right panel) of the interactions between a dummy for grid cells with gold suitability and year fixed effects with their 95% confidence intervals. The omitted year is 2001. The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. Regressions include grid cell and region $\times$ year fixed effects as well as a set of time-varying controls, including (i) temperature, temperature squared, precipitation, precipitation squared, and (ii) interactions between time-invariant controls (latitude, longitude, elevation, ruggedness, and the log of 1 + a grid cell's distance to the coast) and the post-KPCS dummy. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation. The dashed vertical line indicates the start of the KPCS in 2002.

Figure 3: Placebo Test: The Effect of the KPCS on the Incidence of Armed Conflicts in Alluvial Gold Suitability Grid Cells

test for heterogeneous treatment effects, we estimate the following model, building on equation (2):

$$Y_{it} = \alpha + \beta_1 \left(SecondaryDiamondSuitability_i \times Post_t \times KP \right) \\ + \beta_1 \left(SecondaryDiamondSuitability_i \times Post_t \times NonKP \right) \\ + \delta_i + \phi_{rt} + \Gamma X_{it} + \sum_{t=1997}^{2013} \Phi \left(C_i \times Year_t \right) + \epsilon_{it}, \quad (3)$$

where KP is an indicator variable equal to 1 if grid cell i is located in a country that was an early KP member and $NonKP$ is an indicator variable equal to 1 if grid cell i is located in a country that did not join the KP or joined later.

The results of this exercise are shown in Figure 4a, with the top panel plotting the estimates when using ACLED and the bottom panel plotting the estimates when using UCDP. Consistent with our hypothesis, we find that the effect of the certification scheme is comparatively larger for grid cells located in African states that were part of the KP from its start. Indeed, only for these grid cells are the estimates statistically significant.

High vs. Low SD Suitability Grid Cells. We now test the hypothesis that the effect of the KPCS is stronger in grid cells with higher secondary diamond suitability, i.e. with a larger area suitable for secondary diamonds and potentially more conflict. We test this hypothesis by comparing secondary diamond suitability grid cells with above median length of river segments into which diamonds may have been carried (high suitability grid cells) to below median river length (low suitability grid cells). The idea is that grid cells with longer river segments are more likely to have (significant) secondary diamond deposits. Our sample's average river length for above-median cells is 103 km versus 33 km for below-median cells. We use the same specification as before but now include triple interactions with an indicator variable that identifies high suitability grid cells (instead of KP) and one that identifies low suitability grid cells (instead of $NonKP$). Figure 4b plots the coefficient estimates. It confirms that the impact of the KPCS is relatively larger for high SD suitability grid cells, albeit the difference is not statistically significant.

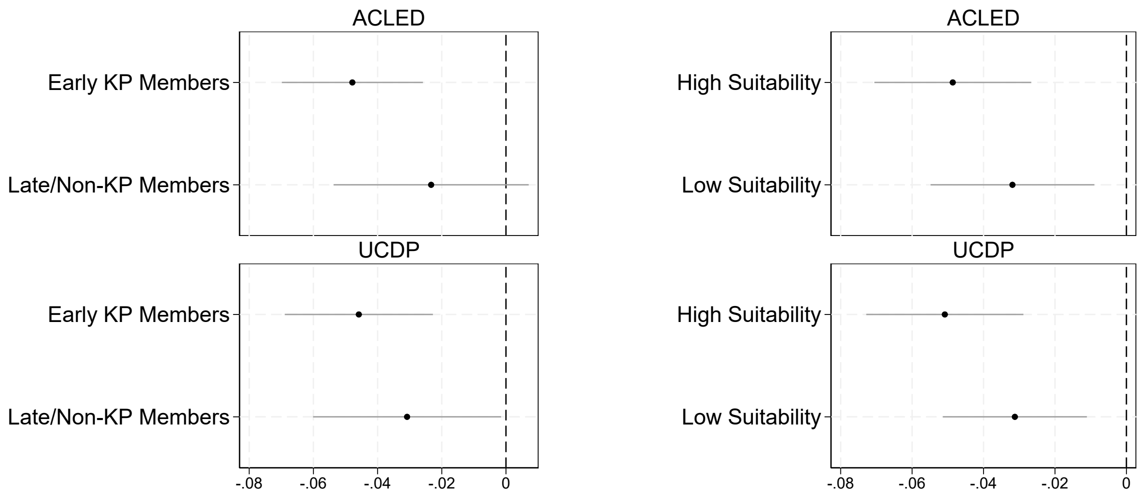
The results of both exercises support a causal relationship between the introduction of the KPCS and armed conflict in Africa.

5.6 Strong vs. Weak Political Institutions

Which countries have benefited the most from the introduction of the KPCS? The Kimberley Process was initiated to address the issue of conflict diamonds – rough diamonds used by rebel groups to finance wars against governments. Implicitly, the KPCS was aimed at supporting, in particular, weak governments. At the same time, an important concern has been that weak political institutions and high levels of corruption undermine the effectiveness of initiatives such as the KPCS. Therefore, this section empirically examines the heterogeneity of treatment effects

(a) Grid Cells in Early vs. Late/ Non-KP Member Countries

(b) Grid Cells with High vs. Low SD Suitability

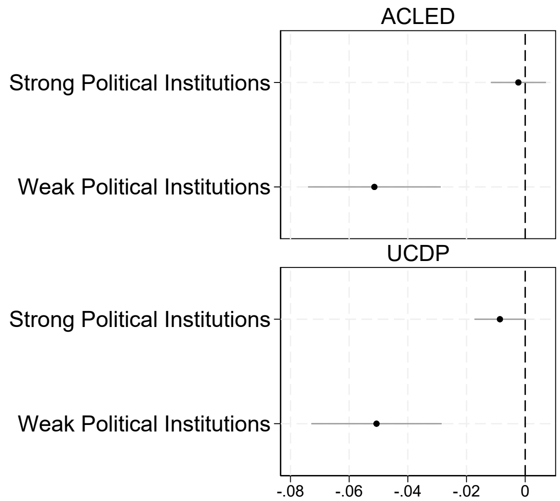


Notes: Panel (a) shows the difference-in-differences estimates for secondary diamond suitability grid cells located in countries that were part of the KPCS from its start vs. those located in countries that did not join immediately and Panel (b) shows difference-in-differences estimates for secondary diamond suitability grid cells with river segments into which diamonds may have been carried with above vs. below median length in km. The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. The horizontal lines indicate the 95% confidence intervals. Regressions are based on equation (2) and include grid cell and region × year fixed effects as well as a set of time-varying controls. The sample is from 1997–2013. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation.

Figure 4: The Effect of the KPCS on Armed Conflict: Heterogeneity by Treatment Intensity

based on the quality of a country's political institutions.

Following previous work, we use the Polity5 Democracy Index as our main source to classify countries as having strong or weak political institutions (Berman et al., 2017; Baik et al., 2024; Christensen et al., 2024). The Polity5 scale ranges from -10 ("hereditary monarchy") to +10 ("consolidated democracy"). We define countries classified as "democracies" (i.e. countries with a Polity5 score of +6 or higher) as having strong political institutions and all other countries as having weak political institutions. Taking the last year (2001) prior to the introduction of the KPCS as a reference year leaves us with eight countries with strong political institutions: Botswana, Ghana, Lesotho, Madagascar, Namibia, Niger, Senegal, and South Africa. To examine the robustness of our results, we draw alternatively on the Worldwide Governance Indicators (WGI) from the World Bank. The WGI are standardized with a mean of zero and a standard deviation of one, with values ranging from -2.5 to +2.5. We define countries with a value of zero or higher as having strong political institutions, while all other



Notes: The figure shows the difference-in-differences estimates for secondary diamond suitability grid cells located in countries classified as having strong political institutions vs. those located in countries with weak political institutions. Countries are classified as having strong political institutions if in the year 2001, they had a Polity5 score of +6 or higher. The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. The horizontal lines indicate the 95% confidence intervals. Regressions are based on equation 3 and include grid cell and region × year fixed effects as well as a set of time-varying controls. The sample is from 1997–2013. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation.

Figure 5: The Effect of the KPCS on Armed Conflict: Heterogeneous Treatment Effects by a Country's Political Institutions

countries are defined as having weak political institutions.[16]

We then estimate equation 3, where the triple interactions are now with an indicator variable that identifies grid cells located in a country defined as having strong political institutions (instead of KP) and an indicator variable that identifies grid cells located in a country with weak political institutions (instead of $NonKP$).

The results are shown in Figure 5. The estimates suggest that countries with weak political institutions drive the decline in armed conflict after 2001. Appendix Figure A6 shows a similar pattern when using the WGI to classify countries. Our results are thus in line with the findings in Christensen et al. (2024) who document that the impact of increased enforcement of the US Foreign Corrupt Practices Act was concentrated in countries with weak political institutions. It is also consistent with theoretical work on the resource curse, which highlights the important role of institutions (e.g. Mehlum et al., 2006; Robinson et al., 2006).

6 Mechanism: Disrupting Rebel Groups' Funding Strategies

Our findings are consistent with the KPCS's core idea of breaking the link between natural resources and conflict. Rebel groups exploit natural resources, generating revenue through

[16] According to this definition, seven countries are classified as having strong political institutions: Botswana, Cape Verde, Mauritius, Namibia, Sao Tome and Principe, Seychelles, and South Africa.

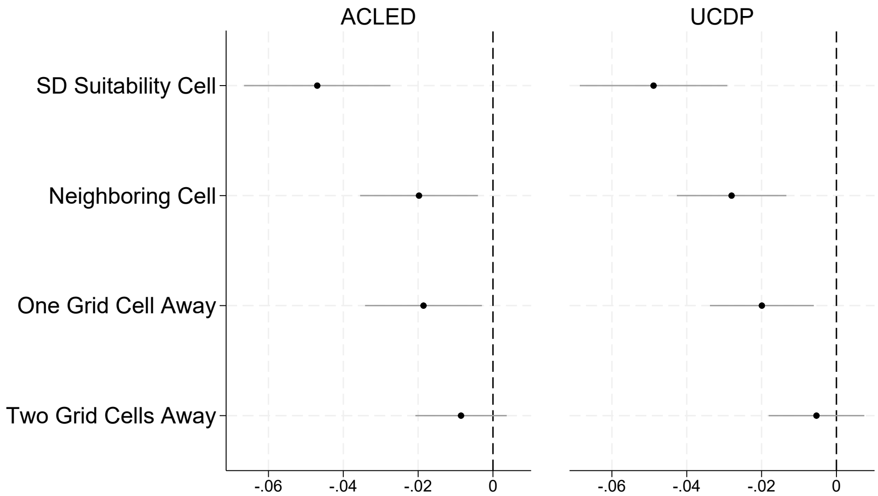
looting, extortion, bribery, illicit trade, or direct control of resource-rich territories (Berman et al., 2017). The barriers to illicit diamond trading imposed by the KPCS may thus have complicated rebel groups' ability to fund armed operations and logistics, restricting their operational feasibility. Consequently, rebels may have relocated their activities to other areas, exploited alternative minerals, or engaged in different illegal activities, potentially shifting violence beyond SD suitability grid cells.

We examine the response of rebel groups using two complementary approaches. First, we investigate whether the decline in armed conflict in SD suitability grid cells alters the incidence of armed conflict and violence more broadly in adjacent grid cells. Next, we use detailed rebel group-level data from ACLED to trace and analyze how rebel group operations and tactics may have changed in response to KPCS-related constraints. The advantage of this analysis is that it captures cases in which rebel groups relocate their violent activities to areas further away. To account for the fact that rebel groups may have engaged in different forms of violence following the introduction of the KPCS due to the need to identify new funding sources, we consider in the following both battles and any violence, which additionally includes violence against civilians and explosions/remote violence.

6.1 Geographic Spillovers

We first provide evidence that the certification scheme reduced armed conflict not only in secondary diamond suitability grid cells, but also in neighboring grid cells. Ex-ante, there may be positive or negative spillovers. On the one hand, rebel groups may relocate their activities, increasing armed conflict in neighboring cells (negative). On the other hand, shutting down a lucrative income source for rebel groups may also reduce their ability to engage in armed conflict, leading to a decline in armed conflict in neighboring cells (positive). We test for geographic spillovers by adding to model (2) interactions between the post-KPCS dummy and dummies for grid cell i having the border of a secondary diamond suitability grid cell within a certain distance to its centroid: (0 degree – 0.25 degrees], (0.25 degrees – 0.75 degrees], and (0.75 degrees – 1.25 degrees]. As the grid cells in our sample are of the size 0.5 $\times$ 0.5 degrees, the first category of grid cells essentially includes immediate neighbors of a secondary diamond suitability cell; the second (third) category of grid cells includes grid cells that are located approximately one (two) grid cells away from a secondary diamond suitability grid cell.

The results are shown in Figure 6. The main treatment effect – the estimate on the interaction between a SD suitability grid cell and the post-KPCS dummy – becomes somewhat larger, which we would expect in the presence of spillovers. The estimates for the new interaction terms suggest positive spillovers (significant at the 1% level) for grid cells located in proximity (up to 0.75 degrees) to an SD suitability grid cell. The estimated spillovers are substantial: grid cells located next to an SD suitability grid cell see a reduction in armed conflict that is about half the size of that for treated cells (-0.020 vs. -0.047 for treated cells based on ACLED; -0.028 vs. -0.049 for treated cells based on UCDP). The estimates for grid cells located one grid cell



Notes: Difference-in-differences estimates on the interactions between a dummy for grid cell i having secondary diamond suitability and the post-KPCS dummy and between dummies for grid cell i having the border of a secondary diamond suitability grid cell within a certain distance from its centroid – (0 degree – 0.25 degrees], (0.25 degrees – 0.75 degrees], and (0.75 degrees – 1.25 degrees] – and the $PostKPCS$ dummy. The horizontal lines indicate the 95% confidence intervals. The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. Regressions are based on equation (2) and include grid cell and region $\times$ year fixed effects as well as a set of time-varying controls. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation. The sample is from 1997–2013.

Figure 6: The Effect of the KPCS on Armed Conflict: Geographic Spillovers

away are of similar size in the case of ACLED (-0.019) and slightly smaller in the case of UCDP (-0.020). Spillovers wear off for grid cells located further away. The positive spillovers imply that the estimated effect of the KPCS on armed conflict based on our baseline specification can be seen as a lower bound.

Using ACLED, Section 5.3 above documents that after the introduction of the KPCS, other forms of violence also declined, suggesting that rebel groups did not resort to alternative forms of violence. To rule out the possibility that rebel groups may have diverted other forms of violent activities to neighboring grid cells, Appendix Figure A7 considers all violent events, not just armed conflict. We observe a pattern comparable to that for armed conflict, indicating that total violence does not increase in grid cells near SD-suitable areas.

6.2 Rebel Group-Level Analysis

Positive spillovers suggest that the KPCS made it more difficult for rebel groups to finance their armed activities. We now add a rebel group-level analysis to provide further evidence that the KPCS significantly reduced the ability (or, feasibility) of rebel groups to initiate, sustain, and escalate violent activities. We use ACLED data, as it provides information on the number and location of violent events in which a particular rebel group was engaged in a given year. This allows us to examine changes in violent activities at the rebel group level as well as changes in the geographic spread of their activities over time.

For all groups for which ACLED reports at least five battle events over the 1997–2013 period, we manually code groups as rebel and non-rebel groups. We then consider all rebel groups that existed prior to the introduction of the KPCS, i.e. for which ACLED reports at least one armed conflict over the 1997–2001 period. We then compare the incidence of armed conflict (or the number of battle events and the area covered) between rebel groups that, prior to the KPCS (over the 1997–2001 period), were involved in armed conflict in at least one grid cell with secondary diamond suitability and rebel groups that did not operate in such an area. Prior to the introduction of the KPCS, 47 (22%) of the 216 rebel groups were involved in some armed conflict in a grid cell with secondary diamond suitability.

More formally, we estimate the following event-study difference-in-differences model:

$$Y_{it} = \alpha + \sum_{t=1997}^{2013} \beta_t \left(PastActivityinDiamondArea_i \times Year_t \right) + \delta_i + \phi_{rt} + \Gamma X_{it} + \sum_{t=1997}^{2013} \Phi \left(C_i \times Year_t \right) + \epsilon_{it} \quad (4)$$

where $Y_{i,t}$ is one of the following outcomes: (1) an indicator variable equal to one if ACLED reports at least one battle event for rebel group i in year t, (2) the natural log of (1+ the number of battle events of rebel group i in year t), and (3) the natural log of (1+ the number of grid cells in which rebel group i engaged in battles in year t). $PastActivityinDiamondArea_i$ is an indicator variable equal to one if rebel group i was involved in at least one battle event between 1997 and 2001 in a grid cell with secondary diamond suitability. We include rebel group fixed effects (δ_i) and region $\times$ year fixed effects (ϕ_{rt}). Moreover, we include the same set of time-varying controls as before. However, since the analysis is now at the level of the rebel groups, which are not tied to a specific grid cell, we do the following. We first identify all the grid cells in which a rebel group participated in an armed conflict over the entire 1997–2013 period.[17] For each time-varying variable (X_{it}) – temperature, temperature squared, precipitation, precipitation squared – we then compute the mean value for the identified (fixed) set of grid cells for each year. For each time-constant variable (C_i) – latitude, longitude, elevation, ruggedness, and the log of (1 + a grid cell's distance to the coast) – we calculate the mean for one year and then interact each mean value with the year dummies. As before, we compute Conley (1999) standard arrows with a distance cut-off of 500 km and infinite serial correlation.

We additionally estimate the following simple difference-in-differences model:

[17] Due to endogeneity concerns regarding the location of rebel groups, we do not consider the spread of rebel groups on an annual basis. In addition, rebel groups do not necessarily engage in armed conflict every year, so years without an armed conflict would have a missing value.

$$Y_{it} = \alpha + \beta\left(PastActivityinDiamondArea_i \times Post_t\right) + \delta_i + \phi_{rt} + \Gamma X_{it} + \sum_{t=1997}^{2013} \Phi\left(C_i \times Year_t\right) + \epsilon_{it} \quad (5)$$

where $Post_t$ is an indicator variable equal to one for the years 2002 to 2013, and zero otherwise. All other variables are defined as before.

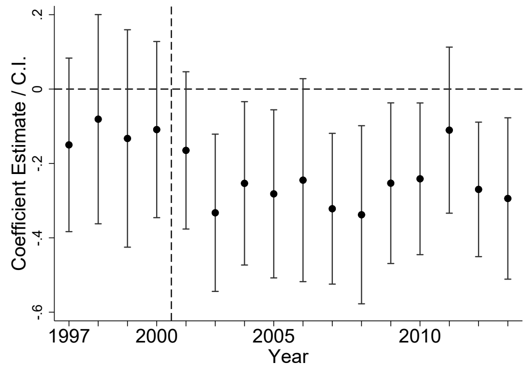
The event-study estimates are plotted in Figure 7. They provide little evidence for a pre-trend. Starting in 2002, there is a differential decline in the probability that rebel groups engage in battles, the number of battles they engage in, and the number of grid cells in which these battle events occur. These results suggest a decrease in violence and a more geographically concentrated pattern of rebel groups' activities after the introduction of the KPCS.

Columns (1) to (3) of Panel (a) of Table 4 report the results from estimating equation (5). We find that rebel groups are less likely to engage in armed conflicts following the introduction of the KPCS (column 1). The effect is significant: while 59% of rebel groups active in an SD suitability grid cell were involved in some armed activity annually prior to the introduction of the KPCS (Appendix Table A7), this number fell by almost 30% under the certification scheme. This decline in the extensive margin is accompanied by a decline in the intensive margin, as the number of conflict events also falls (column 2). The estimate of -0.70 suggests that following the introduction of the KPCS, the number of battle events declined by 50%, from previously 26 battle events annually to about 18 battle events.[18] As rebel groups engage in fewer armed activities after the introduction of the KPCS, we would expect their activities to become more geographically concentrated. This is confirmed in column (3), which considers the number of grid cells in which armed conflict was recorded. Taking the estimate of -0.46 at face value implies that the number of grid cells declined by 39%. This suggests that the area covered by rebel groups decreased from an average of almost six grid cells prior to the introduction of the KPCS to approximately four grid cells after its introduction.

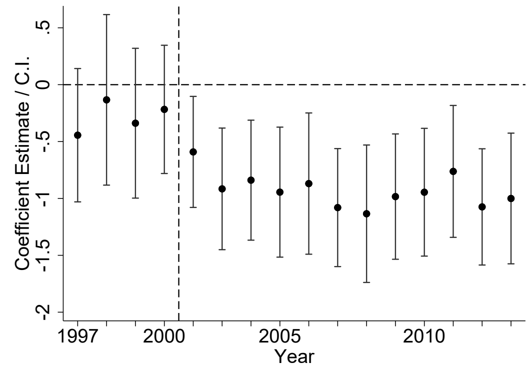
In Panel (b) of Table 4, we find that these results hold when we consider only battle events outside of secondary diamond suitability grid cells. This provides evidence that, in line with the positive spillovers we documented above, rebel groups do not move their activities to other areas. Aside from relocating armed activities to areas without secondary diamond suitability, rebel groups may resort to other forms of violence to secure or develop new sources of income. In Panel (c), we therefore consider any violence (battles, violence against civilians, and explosions/distant violence). Reassuringly, we also observe a decline in overall violence. This provides further evidence that the KPCS may have successfully removed a key financing source for these rebel groups, thereby reducing their ability to engage in armed conflict or other types of violence.

[18] Following Halvorsen and Palmquist (1980), we compute the percent change using the formula $100*[exp(\beta)-1]$.

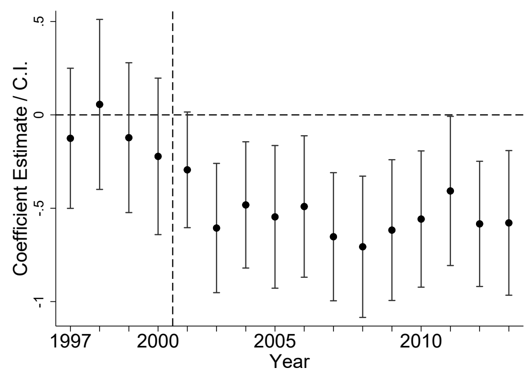
(a) Any Armed Conflict



(b) Ln(1 + Nr Armed Conflicts)



(c) Ln(1 + Nr Grid Cells)



Notes: Difference-in-differences estimates of the interactions between a dummy for rebel groups active in secondary diamond suitability grid cells pre-KPCS and year fixed effects with their 95% confidence intervals indicated by the vertical lines. The dependent variable is any battle event for rebel group i in year t in Figure 7a, the log of 1+ the number of battle events of rebel group i in year t in Figure 7b, and the log of 1+ the number of grid cells in which rebel group i engaged in battles in year t in Figure 7c. Based on ACLED (1997–2013). Restricted to rebel groups involved in at least one armed conflict event prior to KPCS and at least 5 events over the 1997–2013 period (N=216). The omitted year is 2001. The regressions include grid cell and region $\times$ year fixed effects. Standard errors allow for spatial correlation within a radius of 500 km and infinite serial correlation. The dashed vertical line indicates the start of the KPCS in 2002.

Figure 7: The Effect of the KPCS on Rebel Groups' Armed Activities

7 Conclusion

We study the impact of the Kimberley Process Certification Scheme (KPCS) on armed conflict in Africa. To identify its causal effect, we compare grid cells with and without secondary diamond (SD) suitability over time in a difference-in-differences framework. We find that the KPCS significantly and permanently reduced the incidence of armed conflict, controlling for grid cell fixed effects, region $\times$ year fixed effects, and a set of time-varying controls. More specifically, our estimates suggest that in the first 12 years, the KPCS led to about a halving of

Table 4: The Effect of the KPCS on Rebel Groups' Armed Activities

	Any Event	*Ln(1+ ...)*		*Any Event*	*Ln(1+ ...)*	
		Nr Events	*Nr Cells*		*Nr Events*	*Nr Cells*
	(1)	(2)	(3)	(4)	(5)	(6)
	a. All Armed Conflicts			*b. Armed Conflicts in Non-SD Suitability Grid Cells*		
Past Activity in SD Suitability Grid Cell X Post	-0.164***	-0.702***	-0.461***	-0.114**	-0.240***	-0.320***
	(0.054)	(0.169)	(0.123)	(0.057)	(0.040)	(0.118)
Mean of Dep. Var. pre-KPCS	0.591	25.8	5.63	0.494	15.4	4.04
Observations	3,672	3,672	3,672	3,672	3,672	3,672
R^2	0.145	0.184	0.174	0.139	0.959	0.161
Rebel-Group Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Region × Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Time-Varying Controls	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Difference-in-differences estimates on the interactions between a dummy for rebel groups active in secondary diamond suitability grid cells pre-KPCS and the $PostKPCS_t$ dummy. Panel (a) considers battles (armed conflict), Panel (b) considers violence against civilians (based on ACLED). The dependent variable is any event for rebel group i and year t (columns 1 and 3), the log of 1+ the number of events of rebel group i and year t (columns 2 and 5), and the log of 1+ the number of grid cells in which rebel group i engaged in violent events in year t (columns 3 and 6). The regressions include grid cell and region × year fixed effects. Columns 4 to 6 consider only battles that occurred in non-secondary diamond (SD) suitability grid cells. The sample is from 1997 to 2013. 'Mean of Dep. Var. (pre-KPCS)' gives the mean of the dependent variable for SD suitability grid cells over the 1997–2001 period. In columns 2-3 and 5-6, it gives the mean value for the non-transformed variable. Standard errors allow for spatial correlation within a radius of 500 km and infinite serial correlation.

the frequency of armed conflict that occurred in grid cells with secondary diamond suitability prior to the KPCS. This result is robust to a number of alternative specifications. We also show that the KPCS led to a reduction in total violent conflicts in these areas. In addition, we exploit heterogeneity in treatment intensity to provide further evidence of the KPCS's causal effect. We show that the effect is larger for secondary diamond suitability grid cells located in countries that joined the Kimberley Process from its start and for grid cells with high secondary diamond suitability. Finally, we demonstrate that countries with weak political institutions are indeed the main beneficiaries of the KPCS, as intended.

We also present evidence that the KPCS effectively prevents rebel groups from accessing financial resources. First, we document that armed conflict and violence in general also declined in grid cells neighboring SD suitability grid cells. These positive spillovers suggest that conflict is not being displaced into other areas in the form of battles or other types of violence. Second, we examine changes in armed conflict and other violent activities of rebel groups using ACLED's rebel group-level dataset. To this end, we compare rebel groups that, prior to the KPCS, were involved in armed conflict in at least one grid cell with SD suitability and rebel groups that did not operate in such an area. We find that after the introduction of the KPCS, the former group engaged in fewer violent conflicts, and their violent activities became more geographically restricted. Overall, these findings suggest that the KPCS has been successful in cutting off rebel groups' financial resources, thereby reducing both armed conflict and total violent conflict.

Global coordination to enhance transparency and traceability is a cornerstone for promoting peace (Rohner, 2024). Studying one of the first global initiatives – the KPCS – we demonstrate its potential to curb conflict even in settings with weak governance. One possible reason for the effectiveness of the KPCS is that its strict enforcement may not always be essential, as long as governments and other stakeholders are committed to cutting off rebel groups from their sources of funding. Indeed, qualitative research in Sierra Leone shows how various activities, while undermining some regulations of the KPCS, nevertheless prevented rebel groups from profiting from illicit diamonds (Engwicht, 2018). It remains an open question whether initiatives that focus on other forms of violence associated with alluvial diamond mining, such as military violence and violations of human rights, can be similarly effective.

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Appendix

Table of Contents

A Further Background on the Diamond Industry

Until 1960, nearly all diamonds worldwide were produced in Africa (Janse, 1995a, 2007). In the following decades, several non-African countries became important producers: Russia (starting in 1960), Australia (starting in 1979), and Canada (starting in 1998) (Janse, 2007). The diamond industry has been considered a cartel with the main player, the company De Beers, being involved in diamond production and trade since 1880 (Spar, 2006; Haufler, 2009). In the early 1990s, De Beers was responsible for nearly 50% of the world's rough diamond production and sold about 80% of the world's total supply through its Central Trading Organization, now known as the Diamond Trading Company (The Economist, 2007). As Spar (2006) documents in greater detail, De Beers has been highly successful in controlling production and the available stock of rough diamonds on the market. At the same time, De Beers invested in creating sustained demand and a high willingness to pay among consumers by creating the illusion that diamonds are scarce and a symbol of enduring love. Its slogan, "A diamond is forever" became famous. In the 1990s, with the end of the Soviet Union and the end of the apartheid regime in South Africa, the opening of new mines, and antitrust litigation in the US, among others, De Beers faced difficulties in maintaining its system of managing production and demand, and, in turn, its market share declined (Spar, 2006; Haufler, 2009). Additionally, growing international awareness about "blood diamonds" threatened the diamond industry, and De Beers in particular. Some, therefore, argue that De Beers highly benefited from the Kimberley Process as it helped regulate the market and reduce the flow of illegal diamonds into the world market (Spar, 2006; Haufler, 2009).

Before KPCS, data on diamond production, especially alluvial diamond production, were often unreliable if available at all. One indication that production figures are not reliable is the (sometimes vast) discrepancy between export and import values for producing and trading countries (US GAO, 2002). Based on estimated export values for rough diamonds for the year 2000, major producing countries in Africa were (in descending order) Botswana, South Africa, the Democratic Republic of the Congo, Namibia, Angola, the Central African Republic, Guinea, and Liberia (US GAO, 2002, p. 53).

As KP members are required to submit statistics on their diamond production as well as exports and imports, for the most important diamond producers, relevant figures are available from the official KP website (link) as of the year 2004. In Appendix Figure A1 we report figures on the average annual diamond production by volume (carats) and value (US Dollars) for KP members for the period 2004 to 2013. Members with no production are excluded from the table. For members who joined after 2004, we report production averages for the years for which information is available. Although Russia, Australia, and Canada produced significant amounts of diamonds in the 2000s, African countries contributed more than half of all diamonds in the market in terms of both total volume and total value.

Table A1: KP Members' Average Diamond Production (2004-2013)

Country	Carats		Value	
	in Thousands	% of Total	in Millions of $	% of Total
Russian Federation	36,801.15	24.99%	2,583.08	21.78%
Botswana	26,953,26	18.31%	2,941.67	24.81%
Democratic Republic of the Congo	25,186.01	17.10%	328.18	2.77%
Australia	17,088.50	11.61%	354.74	2.99%
Canada	12,467.79	8.47%	1,866.54	15.74%
South Africa	10,996.61	7.47%	1,209.10	10.20%
Angola	8,387.08	5.70%	1,070.87	9.03%
Zimbabwe	4,320.43	2.93%	217.10	1.83%
Namibia	1,843.53	1.25%	778.64	6.57%
Guinea	765.73	0.52%	39.43	0.33%
Ghana	585.82	0.40%	18.35	0.15%
Sierra Leone	528.46	0.36%	129.06	1.09%
Central African Republic	337.99	0.23%	51.28	0.43%
Lesotho	199.74	0.14%	179.40	1.51%
Guyana	192.65	0.13%	24.09	0.20%
Tanzania	191.23	0.13%	26.72	0.23%
Brazil	114.39	0.08%	8.99	0.08%
Congo, Republic of	109.41	0.07%	2.90	0.02%
China, People's Republic of	41.70	0.03%	0.68	0.01%
Liberia	36.93	0.03%	13.10	0.11%
India	31.54	0.02%	4.91	0.04%
Indonesia	25.63	0.02%	5.80	0.05%
Togo	19.59	0.01%	1.91	0.02%
Venezuela	18.00	0.01%	1.22	0.01%
Cameroon	1.76	0.00%	0.42	0.00%
Total	147,244.94	100.0%	11,858.17	100.0%

Notes: This table reports the average annual diamond production by volume (carats) and value (US Dollars) for KP members for the period 2004 to 2013 based on statistics published on the official KP website (link). Members with no production are not included in the table. For members who joined after 2004, we report production averages for years with available information.

B African Countries: KP Membership and Diamond Deposits

Table A2: KP Membership and Diamond Deposits (African Countries)

Country	KP Member? (Year joining)	Primary Diamond Deposits	Secondary Diamond Deposits
Algeria	No	No	Yes
Angola	Yes (founding member)	Yes	Yes
Benin	No	No	No
Botswana	Yes (founding member)	Yes	Yes
Burkina Faso	Applicant	Yes	No
Burundi	No	No	No
Cameroon	Yes (2012)	No	No
Cape Verde	No	No	No
Central African Republic	Yes (founding member)[1]	No	Yes
Chad	No	No	Yes
Comoros	No	No	No
Democratic Republic of the Congo	Yes (founding member)	Yes	Yes
Djibouti	No	No	No
Egypt	No	No	No
Equatorial Guinea	No	No	No
Eritrea	No	No	No
Eswatini	Yes(2011)	Yes	No
Ethiopia	No	No	No
Gabon	2018	Yes	Yes
Gambia	No	No	No
Ghana	Yes (founding member)	No	Yes
Guinea	Yes (founding member)	Yes	Yes
Guinea-Bissau	No	No	No
Ivory Coast	Yes (founding member)	Yes	Yes
Kenya	Applicant	No	No
Lesotho	Yes (founding member)	Yes	Yes
Liberia	Yes (2007)	Yes	Yes
Libya	No	No	No
Madagascar	No	No	No
Malawi	No	No	No
Mali	Yes (2013)	Yes	Yes

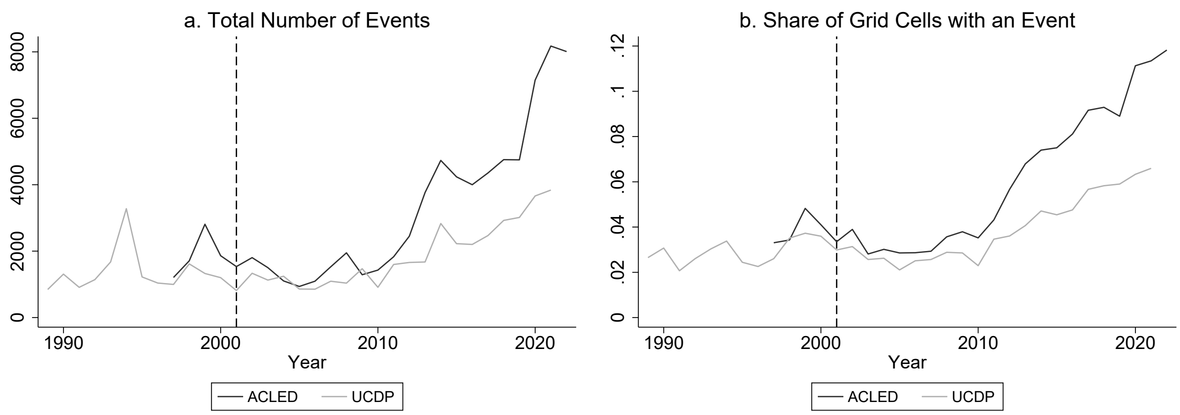
Continued on next page...

Table A3: KP Membership and Diamond Deposits (African Countries) (Cont'd)

Country	KP Member? (Year joining)	Primary Diamond Deposits	Secondary Diamond Deposits
Continued from previous page...			
Mauritania	Applicant	Yes	No
Mauritius	Yes (founding member)	No	No
Morocco	No	No	No
Mozambique	Yes (2021)	Yes	Yes
Namibia	Yes (founding member)	Yes	Yes
Niger	No	No	No
Nigeria	No	No	Yes
Republic of the Congo	Yes (founding member/2007)[2]	No	Yes
Rwanda	No	No	No
Sahrawi Arab Democratic Republic	No	No	No
São Tomé and Príncipe	No	No	No
Senegal	No	No	No
Seychelles	No	No	No
Sierra Leone	Yes (founding member)	Yes	Yes
Somalia	No	No	No
South Africa	Yes (founding member)	Yes	Yes
South Sudan	No	No	No
Sudan	No	No	No
Tanzania	Yes (founding member)	Yes	Yes
Togo	Yes (founding member)	No	No
Tunisia	No	No	No
Uganda	No	No	No
Zambia	Applicant	No	Yes
Zimbabwe	Yes (founding member)	Yes	Yes

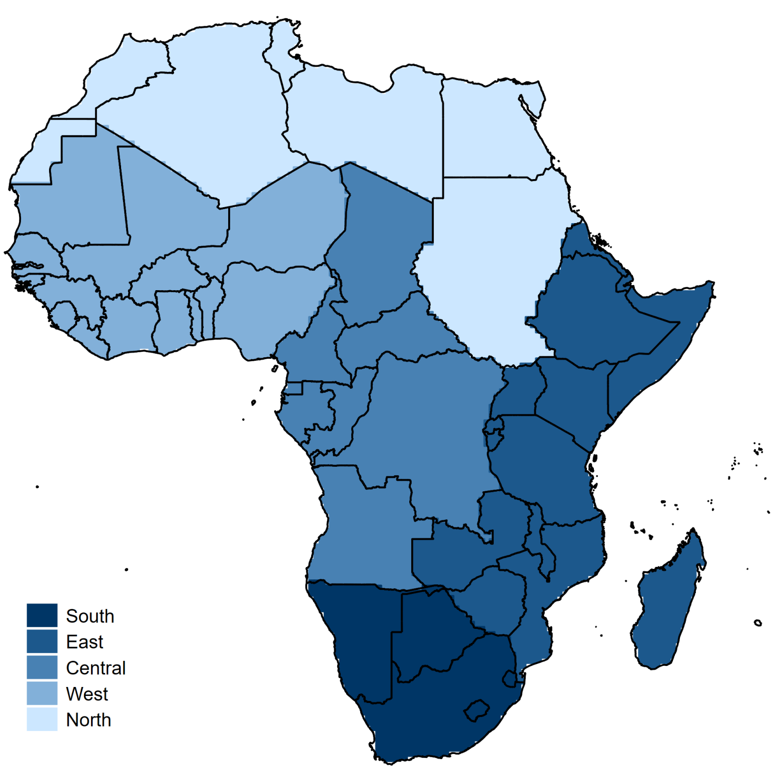
Notes: In 2003, 16 African states were members of the Kimberley Process (KP). Other early members were, among others, Australia, Brazil, Canada, China, the European Union (counted as a single participant), India, Israel, Japan, Norway, Russia, Switzerland, and the United States.
[1] The Central African Republic was temporary suspended from 2013-2015.
[2] The Republic of the Congo was expelled in 2004 and readmitted in 2007.
Sources: Column 2: KP website (link), columns 3 and 4: Gilmore et al. (2005).

C Descriptive Statistics



Notes: This figure shows in Panel (a) the total number of conflict events by year and in Panel (b) the share of grid cells with at least one conflict event by year. This is shown for battles reported by ACLED (dark grey line) and for armed conflicts reported by UCDP (light grey line).

Figure A1: Armed Conflict Events Over Time: ACLED and UCDP



Notes: Following Christensen et al. (2024), we include region × year fixed effects in our baseline specification, see Section 4. This map illustrates the five regions, each shown in a different shade.

Figure A2: Map of African Regions

Table A4: Descriptive Statistics (Grid Cell-Level)

Incidence of Armed Conflict	Observations	Mean	Std. Dev.	Min	Max
Based on ACLED:					
Prior to the Introduction of the KPCS (1997-2001):					
All cells	50,625	0.038	0.192	0	1
Secondary diamond suitability =1	3,445	0.087	0.282	0	1
Secondary diamond suitability=0	47,180	0.035	0.183	0	1
After the Introduction of the KPCS (2002-2013):					
All cells	121,500	0.038	0.191	0	1
Secondary diamond suitability=1	8,268	0.036	0.186	0	1
Secondary diamond suitability=0	113,232	0.038	0.191	0	1
Based on UCDP:					
Prior to the Introduction of the KPCS (1997-2001):					
All cells	50,625	0.032	0.177	0	1
Secondary diamond suitability=1	3,445	0.070	0.255	0	1
Secondary diamond suitability=0	47,180	0.030	0.170	0	1
After the Introduction of the KPCS (2002-2013):					
All cells	121,500	0.029	0.167	0	1
Secondary diamond suitability=1	8,268	0.021	0.144	0	1
Secondary diamond suitability=0	113,232	0.029	0.168	0	1

Notes: Details on the variables are provided in Section 3.

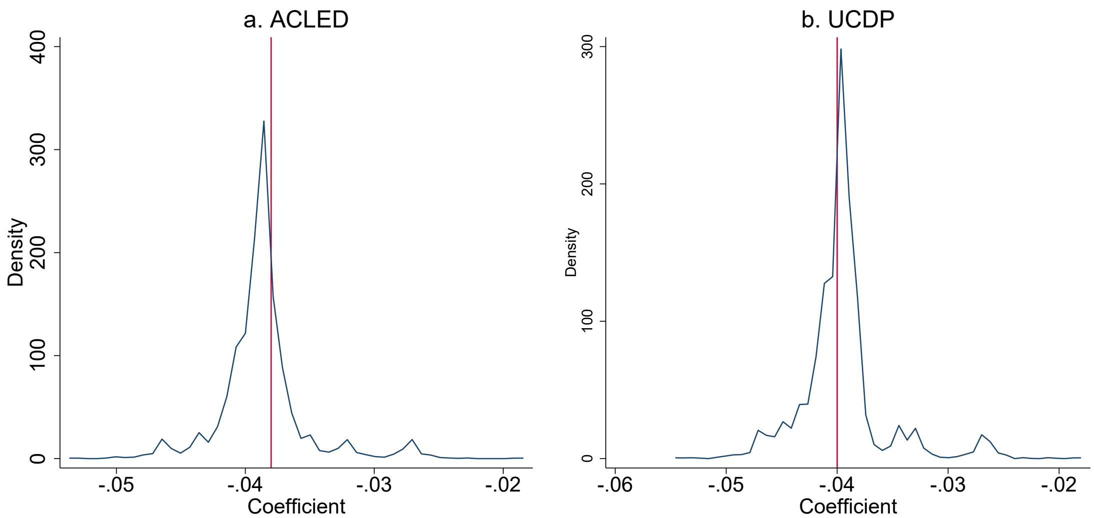
D Robustness Analyses

Table A5: The Effect of the Kimberley Process on Armed Conflicts: Country x Year FE

	Incidence of Armed Conflict						
	Baseline					River Cells	Gold Cells
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: ACLED							
SD Suitability × Post	-0.024***	-0.012**	-0.015*	-0.023***	-0.031***	-0.013	-0.018*
	(0.008)	(0.006)	(0.008)	(0.008)	(0.009)	(0.008)	(0.010)
Mean Incidence pre-KPCS	0.087	0.054	0.082	0.087	0.087	0.094	0.072
Observations	172,125	164,662	169,099	172,125	172,125	60,571	74,885
R^2	0.319	0.326	0.319	0.319	0.319	0.330	0.322
Panel B: UCDP							
SD Suitability × Post	-0.027***	-0.010*	-0.025***	-0.027***	-0.028***	-0.024***	-0.018**
	(0.008)	(0.006)	(0.009)	(0.008)	(0.010)	(0.009)	(0.008)
Mean Incidence pre-KPCS	0.070	0.038	0.072	0.070	0.070	0.077	0.058
Observations	172,125	164,662	169,099	172,125	172,125	60,571	74,885
R^2	0.323	0.330	0.323	0.324	0.323	0.300	0.301
Grid-Cell FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country × Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time-Varying Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
W/o Angola & Sierra Leone	No	Yes	No	No	No	No	No
W/o PD Suitability Cells	No	No	Yes	No	No	No	No
Dist. Ind. Mine × Year FE	No	No	No	Yes	No	No	No
SD Suit. × Mineral Prices	No	No	No	No	Yes	No	No

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Difference-in-differences estimates on the interaction between a dummy for grid cells with secondary diamond (SD) suitability and the post-KPCS dummy. The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. All regressions include grid cell and region × year fixed effects as well as a set of time-varying controls. These controls include (i) temperature, temperature squared, precipitation, precipitation squared, and (ii) interactions between time-invariant controls (latitude, longitude, elevation, ruggedness, and the log of (1 + a grid cell's distance to the coast)) and the post-KPCS dummy. The study period is from 1997-2013 and countries started to implement the KPCS in 2002. 'Mean incidence pre-KPCS' gives the mean armed conflict incidence for secondary diamond suitability grid cells over the 1997–2001 period. Column (1) reports our baseline specification. Column (2) removes grid cells located in Angola and Sierra Leone. Column (3) removes grid cells with primary diamond (PD) suitability. Column (4) adds interactions between the log of (1 + a grid cell's distance to an industrial diamond mine (pre-KPCS)) and year fixed effects. Column (5) adds interactions between the dummy for secondary diamond suitability and the world prices of five important minerals in Africa (diamonds, gold, silver, copper, and tin). Column (6) restricts the sample to grid cells with alluvial gold suitability. Column (7) restricts the sample to grid cells that have a major river within 25km from their border. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation.



Notes: This figure plots the distribution of the estimate on the interaction term between a dummy for grid cells with secondary diamond suitability and the Post-KPCS dummy when re-estimating the baseline specification and removing the grid cells of (i) one country at a time and (ii) two countries at a time. Based on ACLED (left panel) and UCDP (right panel). The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. Regressions include grid cell and region $\times$ year fixed effects as well as a set of time-varying controls. Standard errors allow for spatial correlation within a radius of 500 km and infinite serial correlation.

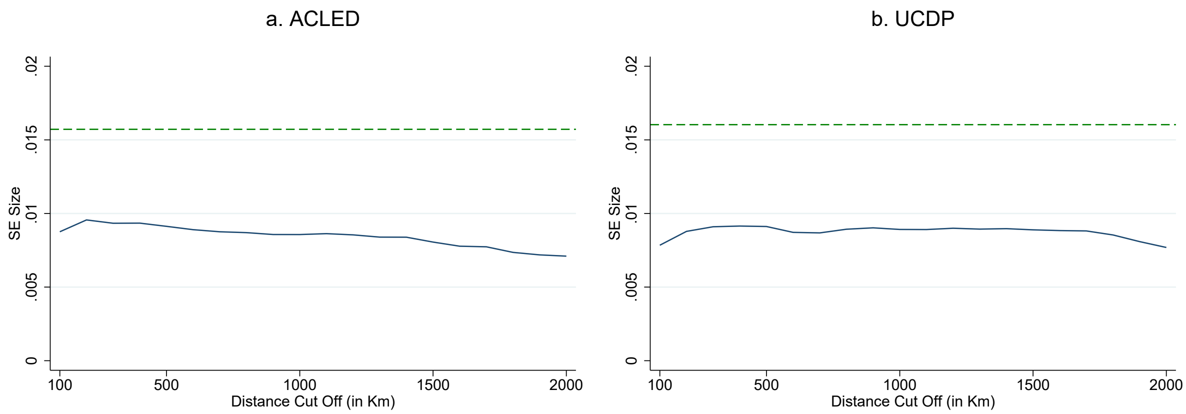
Figure A3: The Effect of the KPCS on the Incidence of Armed Conflicts: Influential Observations

Table A6: The Effect of the Kimberley Process on Armed Conflicts: Dropping Grid Cells from one Region at a Time

	Incidence of Armed Conflict					
	Full Sample	w/o North	w/o West	w/o East	w/o Central	w/o South
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: ACLED (1997–2013)						
Secondary Diamond Suitability × Post	-0.041***	-0.035***	-0.037***	-0.049***	-0.029***	-0.049***
	(0.009)	(0.009)	(0.009)	(0.013)	(0.009)	(0.011)
Mean Conflict Incidence pre-KPCS	0.087	0.085	0.078	0.113	0.058	0.109
Observations	172,125	122,553	137,632	136,833	135,422	156,060
R^2	0.319	0.318	0.325	0.302	0.331	0.319
Panel B: UCDP (1997–2013)						
Secondary Diamond Suitability × Post	-0.041***	-0.037***	-0.039***	-0.052***	-0.027***	-0.050***
	(0.009)	(0.009)	(0.009)	(0.013)	(0.009)	(0.011)
Mean Conflict Incidence pre-KPCS	0.070	0.067	0.062	0.095	0.042	0.088
Observations	172,125	122,553	137,632	136,833	135,422	156,060
R^2	0.323	0.303	0.341	0.310	0.338	0.323
Grid-Cell Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Region × Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Basic Time-Varying Controls	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

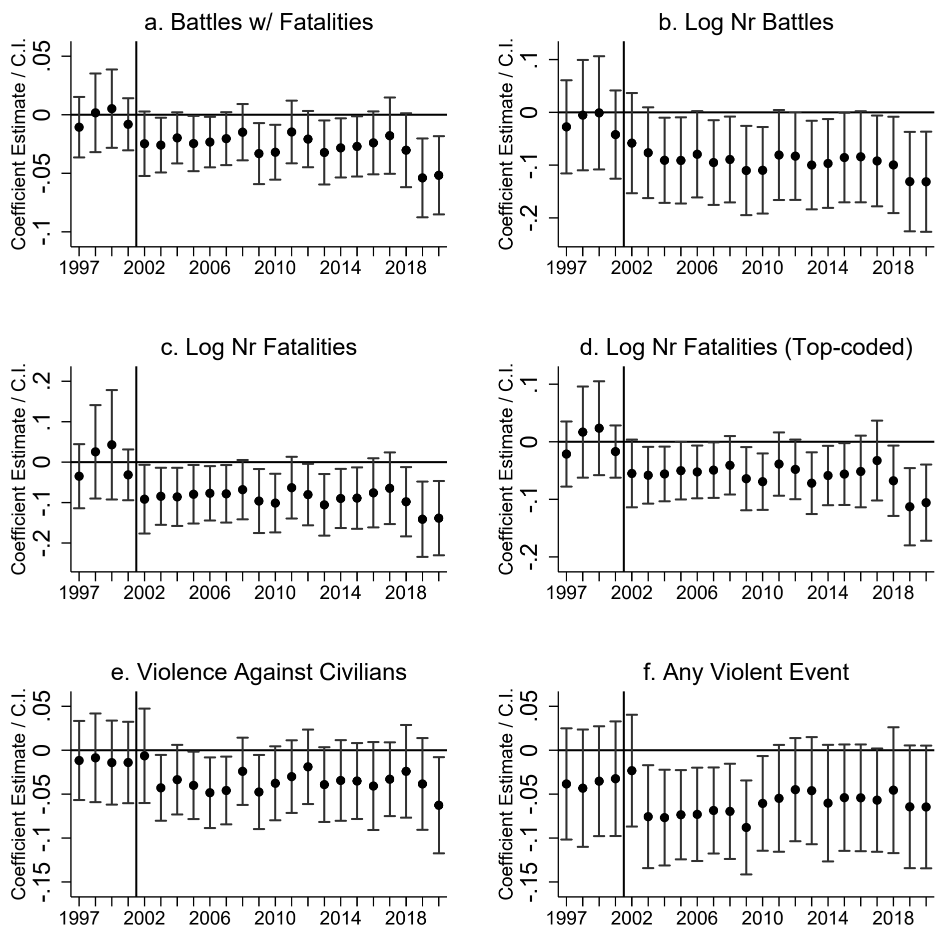
Notes: Difference-in-differences estimates on the interaction between a dummy for grid cells with secondary diamond suitability and the $PostKPCS_t$ dummy. The dependent variable is an indicator variable equal to 1 if at least one armed conflict occurred in grid cell i and year t, and 0 otherwise. 'Mean conflict incidence pre-KPCS' gives the mean conflict incidence for secondary diamond suitability grid cells over the 1997–2001 period. Column (1) reports our baseline specification. In the following columns, we drop all grid cells located in one region. For example, column (2) drops all grid cells located in the northern region. Standard errors allow for spatial correlation within a radius of 500 km and infinite serial correlation.



Notes: This figure examines the sensitivity of our main estimate (column 1 in Panel (a)/(b) of Table 2) to computing Conley (1999) standard errors that allow for spatial correlation within different radii (between 100 km and 2,000 km in steps of 100 km) and infinite serial correlation. In our baseline specification, we use a radius of 500 km. The green lines denote the threshold for significance at the 1% level based on the respective estimate reported in column (1) of Table 2).

Figure A4: The Effect of the Kimberley Process on Armed Conflicts: Different Versions of Conley (1999) standard errors

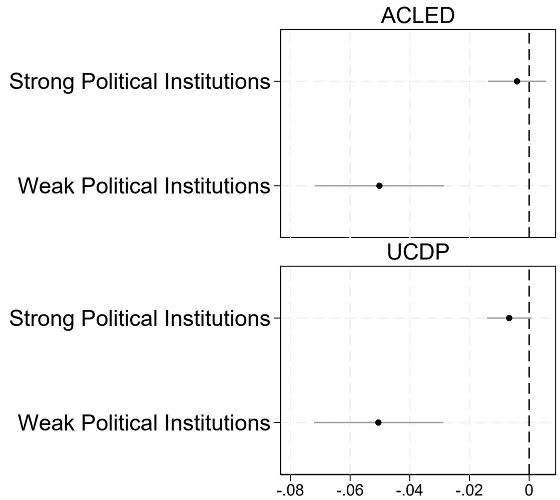
E Alternative Measures for Armed Conflict and Other Forms of Violence



Notes: Difference-in-differences estimates of the interactions between a dummy for grid cells with secondary diamond suitability and year fixed effects with their 95% confidence intervals. The dependent variables is an indicator variable equal to 1 if at least one armed conflict with fatalities occurred in grid cell i and year t, and 0 otherwise (Panel a); the natural log of (1+ the number of battle events in grid cell i and year t) (Panel b); the natural log of (1+ the number of fatalities in grid cell i and year t) (Panel c.); is the same as in Panel c, but top-coded at 14 fatalities (99% percentile) (Panel d); an indicator variable equal to 1 if at least one violent event against civilians occurred in grid cell i and year t, and 0 otherwise (Panel e); and an indicator variable equal to 1 if any violent event occurred in grid cell i and year t, and 0 otherwise (Panel f). The omitted year is 2001. Regressions include grid cell and region $\times$ year fixed effects as well as a set of time-varying controls, including (i) temperature, temperature squared, precipitation, precipitation squared, and (ii) interactions between time-invariant controls (latitude, longitude, elevation, ruggedness, and the log of 1 + a grid cell's distance to the coast) and the post-KPCS dummy. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation. The dashed vertical line indicates the start of the KPCS in 2002.

Figure A5: The Effect of the Kimberley Process on Armed Conflicts Using Alternative Measures and Other Forms of Violence

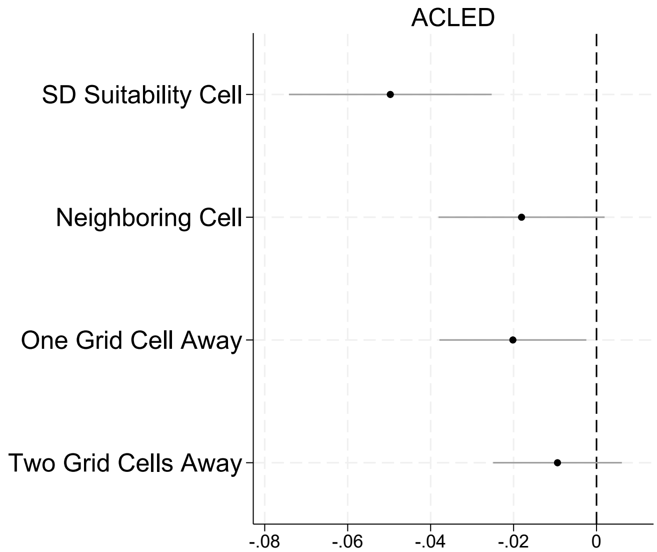
F Strong vs. Weak Political Institutions



Notes: The figure shows the difference-in-differences estimates for secondary diamond suitability grid cells located in countries classified as having strong political institutions vs. those located in countries with weak political institutions. Countries are classified as having strong political institutions if in the year 2001, they had a Worldwide Governance Indicator of zero or higher. The horizontal lines indicate the 95% confidence intervals. Regressions are based on equation 3 and include grid cell and region × year fixed effects as well as a set of time-varying controls. The sample is from 1997–2013. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation.

Figure A6: Heterogeneous Treatment Effects by a Country's Political Institutions: Worldwide Governance Indicators

G Mechanisms



Notes: Difference-in-differences estimates on the interactions between a dummy for grid cell i having secondary diamond suitability and the post-KPCS dummy and between dummies for grid cell i having the border of a secondary diamond suitability grid cell within a certain distance from its centroid – (0 degree – 0.25 degrees], (0.25 degrees – 0.75 degrees], and (0.75 degrees – 1.25 degrees] – and the $PostKPCS$ dummy. The horizontal lines indicate the 95% confidence intervals. The dependent variable is an indicator variable equal to 1 if at least one violent conflict occurred in grid cell i and year t, and 0 otherwise. Regressions are based on equation (2) and include grid cell and region × year fixed effects as well as a set of time-varying controls. We compute Conley standard errors allowing for spatial correlation within a radius of 500 km and infinite serial correlation. The sample is from 1997–2013.

Figure A7: The Effect of the KPCS on Overall Violence: Geographic Spillovers

Table A7: Descriptive Statistics (Rebel Group-Level)

The Incidence of Violent Conflict	Observations	Mean	Std. Dev.	Min	Max
Armed Conflict (All Grid Cells):					
Prior to the Introduction of the KPCS (1997-2001):					
All groups	1,080	0.481	0.500	0	1
Active in diamond cell=1	235	0.591	0.493	0	1
Not active in diamond cell=0	845	0.451	0.498	0	1
After the Introduction of the KPCS (2002-2013):					
All groups	2,592	0.337	0.473	0	1
Active in diamond cell=1	564	0.257	0.437	0	1
Not active in diamond cell=0	2,028	0.359	0.480	0	1
Armed Conflict in Non-SD Suitability Grid Cells:					
Prior to the Introduction of the KPCS (1997-2001):					
All groups	1,080	0.460	0.499	0	1
Active in diamond cell=1	235	0.494	0.501	0	1
Not active in diamond cell=0	845	0.451	0.498	0	1
After the Introduction of the KPCS (2002-2013):					
All groups	2,592	0.324	0.468	0	1
Active in diamond cell=1	564	0.215	0.411	0	1
Not active in diamond cell=0	2,028	0.355	0.479	0	1
Any Violent Event (All Grid Cells):					
Prior to the Introduction of the KPCS (1997-2001):					
All groups	1,080	0.537	0.499	0	1
Active in diamond cell=1	235	0.647	0.479	0	1
Not active in diamond cell=0	845	0.507	0.500	0	1
After the Introduction of the KPCS (2002-2013):					
All groups	2,592	0.409	0.492	0	1
Active in diamond cell=1	564	0.349	0.477	0	1
Not active in diamond cell=0	2,028	0.425	0.494	0	1

Notes: For details on the variables see Section 6.2. SD stands for secondary diamonds. The data comprises of 216 rebel groups, of which 47 (22%) were involved in some armed conflict in a grid cell with secondary diamond suitability prior to the introduction of the KPCS.

Table A8: The Effect of the KPCS on Rebel Groups' Violent Activities

	Any Event	*Ln(1+ ...)*	
		Nr Events	*Nr Cells*
	(1)	(2)	(3)
	Any Violence		
Past Activity in SD Suitability Grid Cell X Post	-0.118***	-0.761***	-0.472***
	(0.054)	(0.192)	(0.133)
Mean of Dep. Var. pre-KPCS	0.647	45.7	7.97
Observations	3,672	3,672	3,672
R^2	0.138	0.181	0.173
Rebel-Group Fixed Effects	Yes	Yes	Yes
Region × Year Fixed Effects	Yes	Yes	Yes
Controls	Yes	Yes	Yes

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Difference-in-differences estimates on the interactions between a dummy for rebel groups active in secondary diamond suitability grid cells pre-KPCS and the $PostKPCS_t$ dummy. All violent conflicts (based on ACLED) are considered, not just armed conflict. The dependent variable is any event for rebel group i and year t (column 1), the log of 1+ the number of events of rebel group i and year t (column 2), and the log of 1+ the number of grid cells in which rebel group i engaged in violent events in year t (column 3). The regressions include grid cell and region × year fixed effects. The sample is from 1997 to 2013. 'Mean of Dep. Var. (pre-KPCS)' gives the mean of the dependent variable for SD suitability grid cells over the 1997–2001 period. Standard errors allow for spatial correlation within a radius of 500 km and infinite serial correlation.